Name

Mathematics Advanced Year 12

Geometric Sequences and Series

Book 3 Geometric sequences and series

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Syllabus Content

MAV-12-03 uses arithmetic and geometric sequences and series to model and solve problems

Geometric sequences and series

- Define a sequence a_n to be a geometric sequence, or geometric progression (GP), if every ratio $\frac{a_n}{a_{n-1}}$ of successive terms is the same where $n \geq 2$, that is $a_n = ra_{n-1}$ for some non-zero real number r called the common ratio
- Develop the formula $a_n = ar^{n-1}$ for the nth term of a GP, where a is the first term and $n \geq 1$, and use it to solve problems
- Recognise that a_n is an exponential function of n in a GP
- Prove by expansion $(x-1)(x^{n-1} + x^{n-2} + \cdots + x^2 + x + 1) = x^n - 1$ for whole numbers $n \geq 1$
- Develop the formula $S_n = \frac{a(1-r^n)}{1-r}$ for the sum of the first n terms of a GP where $r \neq 1$, and use this formula to solve problems
- Examine the behaviour of a_n and S_n as $n \to \infty$ for a GP when $|r| < 1$
- Develop the formula $S_\infty = \frac{a}{1-r}$ for the limiting sum of a geometric series with $|r| < 1$, and use this formula to solve problems
- Apply the formulas for geometric sequences and series to model and solve growth and decay problems where a quantity is an exponential function of time

Geometric Sequences

ⓘ Geometric Sequences

A **geometric sequence**, also called a **geometric progression** or **exponential sequence,** is a sequence where there is a **common ratio** between each pair of consecutive terms.

For example:

$$4, 12, 36, 108, 324 \ldots$$

The common ratio (r) is the same value that you multiply each term to get the next term.
It can be:

- Greater than one (sequence increases, also called exponential growth)
- Between zero and one (sequence decreases, also called exponential decay)
- Negative (sequence alternates between positive and negative terms)

We check whether a sequence is geometric by finding the ratio between consecutive terms and seeing if they are constant.

$$\frac{a_n}{a_{n-1}} = r$$

ⓘ **Identify** whether a sequence is geometric.

✗	✓	✓	✗
1, 4, 9, 16, 25…	1, 4, 16, 64, 256…	150, 75, 37.5, 18.75..	150, 155, 160, 165…

Determine whether the following are geometric sequences.

a	2, 6, 18, 54, 162…	**b**	1000, 100, 10, 1, 0.1…	**c**	$1, -2, 4, -8, 16, -32\ldots$
d	2, 5, 10, 17, 26 ...	**e**	$2,\ 2\sqrt{2},\ 4,\ 4\sqrt{2}\ldots$	**f**	$\frac{1}{2}, \frac{1}{5}, \frac{2}{25} \ldots$

ⓘ a and d

Let the first term of a sequence be a.
Let the common ratio be r.

A geometric sequence is a, ar, ar^2, ar^3 ...

Interpret a and d.

a Write the first five terms of a geometric sequence with $a = 3, r = 2$

b Write the first five terms of a geometric sequence with $a = 45, d = \frac{1}{3}$

c Write the first five terms of a geometric sequence with $a = 2, d = -3$

d Write the first five terms of a geometric sequence with $a = -5, d = 2\pi$

ⓘ **Investigation** General term of any geometric sequence.

We can find the nth term formula of a geometric sequence in terms of a and d:

$a_1 = a$

$a_2 = ar$

$a_3 = ar^2$

$a_4 =$ .....

$a_5 =$ .....

...

$a_n =$ .....

ⓘ n^{th} term Formula for a Geometric Sequence

first term

$$a_n = ar^{n-1}$$

common ratio

Example Finding and applying the rule of a geometric sequence.

Consider the geometric sequence: 2, 6, 18, 54…

a Find the nth term formula for the sequence.

b What is the 12th term?

c Is 1500 a term of this sequence?

a.

$r = 3, a = 2$

$a_n = ar^{n-1}$

$a_n = 2 \times 3^{n-1}$

b.

Substitute $n = 12$ into formula

$a_{12} = 2 \times 3^{11}$

$= 354\,294$

c.

$2 \times 3^{n-1} = 1500$

$3^{n-1} = 750$

750 is not a power of 3, so no.

Alternatively, use logarithms:

$n - 1 = \log_3 750$

$n = \log_3(750) + 1$

$\approx 7.02\ldots$

n must be a whole number, so no.

Practice

Consider the geometric sequence: 5, 10, 20, 40…

a Find the nth term formula for the sequence.

b What is the 10th term?

c Is 320 a term of this sequence?

$a_n = 5 \times 2^{n-1}$

$a_{10} = 2560$

Yes

Consider the geometric sequence: $\frac{2}{3}, \frac{4}{15}, \frac{8}{75}$

a Find the nth term formula for the sequence.

b What is the 8th term? Answer in index form.

$a_n = \frac{2}{3}\left(\frac{2}{5}\right)^{n-1}$

$a_8 = \frac{2^8}{3(5^7)}$

Example Further problems involving geometric sequences.

The 2nd term of a GP is 112 and the 6th term is 7. Find the first term(s).

$a_2 = ar^1 = 112 \quad (1)$

$a_6 = ar^5 = 7 \quad (2)$

$\frac{(2)}{(1)} = r^4 = \frac{1}{16}$

$\therefore r = \pm\frac{1}{2}$

Substituting r into (1) to solve for a:

If $r = \frac{1}{2}, a = 224$

If $r = -\frac{1}{2}, a = -224$

Practice

The 3rd term of a geometric sequence is 18 and the 7th term is 1458.
Find the first term.

2

Example Further problems involving geometric sequences.

Find the first value of n for which a term of the sequence $2, 10, 50$ ... exceeds 3000.

$a = 2, r = 5$

$a_n = ar^{n-1} > 3000$

$2(5)^{n-1} > 3000$

$5^{n-1} > 1500$

$n - 1 > \log_5 1500$

$n > \log_5 1500 + 1$

$n > 5.54 \ldots$ $\therefore$ the 6th term is greater than 3000.

Practice

Find the first value of n for which a term of the sequence $\frac{2}{7}, \frac{6}{7}, \frac{18}{7}$ ... is greater than 100.

2

1 Write out the next three terms of each sequence. They are all GPs.

a 1, 2, 4, ...

8, 16, 32

b 81, 27, 9, ...

$3, 1, \frac{1}{3}$

c −7, −14, −28, ...

−56, −112, −224

d −2500, −500, −100, ...

$-20, -4, -\frac{4}{5}$

e 3, −6, 12, ...

−24, 48, −96

f −25, 50, −100, ...

200, −400, 800

g 5, −5, 5, ...

−5, 5, −5

h −1000, 100, −10, ...

1, −1/10, 1/100

i 0.04, 0.4, 4, ...

40, 400, 4000

2 Write out the first four terms of the GPs whose first terms and common ratios are:

a $a = 12 \quad r = 2$

12, 24, 48, 96

b $a = 5 \quad r = -2$

5, −10, 20, −40

c $a = 18 \quad r = \frac{1}{3}$

$18, 6, 2, \frac{2}{3}$

d $a = 18 \quad r = -1/3$

$18, -6, 2, -\frac{2}{3}$

e $a = 6 \quad r = -1/2$

$6, -3, \frac{3}{2}, -\frac{3}{4}$

f $a = -7 \quad r = -1$

−7, 7, −7, 7

3 Find the ratios $\frac{a_3}{a_2}$ and $\frac{a_3}{a_2}$ for each sequence to test whether it is a GP.

If the sequence is a GP, write down the first term a and the common ratio r.

a $4, 8, 16, \ldots$

$GP: a = 4, r = 2$

b $16, 8, 4, \ldots$

$GP: a = 16, r = \frac{1}{2}$

c $2, 4, 6, \ldots$

not a GP

d $-1000, -100, -10, \ldots$

$GP: a = -1000, r = \frac{1}{10}$

e $-80, 40, -20, \ldots$

$GP: a = -80, r = -\frac{1}{2}$

f $29, 29, 29, \ldots$

$GP: a = 29, r = 1$

g $1, 4, 9, \ldots$

not a GP

h $-14, 14, -14, \ldots$

$GP: a = -14, r = -1$

i $6, 1, \frac{1}{6}, \ldots$

$GP: a = 6, r = \frac{1}{6}$

4 Use the formula $a_n = ar^{n-1}$ to find the fourth term of the GP with:

a $a = 5 \quad r = 2$

40

b $a = 300 \quad r = \frac{1}{10}$

$\frac{3}{10}$

c $a = -7 \quad r = 2$

−56

d $a = -64 \quad r = 1/2$

−8

e $a = 11 \quad r = -2$

−88

f $a = -15 \quad r = -25$

120

5 Use the formula $a_n = ar^{n-1}$ to write the nth term a_n of the GP with:

a $a = 1 \quad r = 3$

$a_n = 3^{n-1}$

b $a = 5 \quad r = 7$

$a_n = 5 \times 7^{n-1}$

c $a = 8 \quad r = -\frac{1}{36}$

$a_n = 8 \times \left(-\frac{1}{3}\right)^{n-1}$

6

a Find the first term a and the common ratio r of the GP 7, 14, 28, ...

a = 7, r = 2

b Find the sixth term a_6 and an expression for the 50th term a_{50}.

$a_6 = ar^5 = 7(2^5) = 224$

$a_{50} = ar^{49} = 7 \times 2^{49}$

c Find a formula for the nth term a_n.

$a_n = ar^{n-1} = 7 \times 2^{n-1}$

7

a Find the first term a and the common ratio r of the GP 10, −30, 90, ...

a = 10, r = −3

b Find the sixth term a_6 and an expression for the 25th term a_{25}.

$a^6 = ar^5 = 10(-3)^5 = -2430$

$a_{25} = ar^{24} = 10(-3)^{24} = 10 \times 3^{24}$

c Find a formula for the nth term a_n.

$a_n = ar^{n-1} = 10 \times (-3)^{n-1}$

8 Find the common ratio of each GP, find a formula for a_n, and find a_6.

a 1, −1, 1, ...

$r = -1$
$a_n = (-1)^{n-1}$
$a_6 = (-1)^5 = -1$

b −2, 4, −8, ...

$r = -2$
$a_n = (-2)^n$
$a_6 = (-2)^6 = 64$

c −8, 24, −72, ...

$r = -3$
$a_n = -8(-3)^{n-1}$
$a_6 = -8(-3)^5 = -8(-243) = 1944$

d 60, −30, 15, ...

$r = -\frac{1}{2}$
$a_n = 60 \times \left(-\frac{1}{2}\right)^{n-1}$
$a_6 = 60\left(-\frac{1}{2}\right)^5 = -\frac{15}{8}$

e −1024, 512, −256, ...

$r = -\frac{1}{2}$
$a_n = 60 \times \left(-\frac{1}{2}\right)^{n-1}$
$a_6 = 60\left(-\frac{1}{2}\right)^5 = -\frac{15}{8}$

f $\frac{1}{16}, -\frac{3}{8}, \frac{9}{4}, \ldots$

$r = -6$
$a_n = \left(\frac{1}{16}\right) \times (-6)^{n-1}$
$a_6 = \left(\frac{1}{16}\right)(-6)^5 = -486$

9 Use the formula $a_n = ar^{n-1}$ to find how many terms there are in each finite sequence.

a $1, 2, 4, \ldots, 64$

$a = 1, r = 2$
$a_n = ar^{n-1}$
$64 = 1(2^{n-1})$
$2^6 = 2^{n-1}$
$n - 1 = 6$
$n = 7$

b $-1, -3, -9, \ldots, -81$

$a = -1, r = 3$
$a_n = ar^{n-1}$
$-81 = -1(3^{n-1})$
$81 = 3^{n-1}$
$3^4 = 3^{n-1}$
$n - 1 = 4$
$n = 5$

c $8, 40, 200, \ldots, 125000$

$a = 8, r = 5$
$a_n = ar^{n-1}$
$125000 = 8(5^{n-1})$
$15625 = 5^{n-1}$
$5^6 = 5^{n-1}$
$n - 1 = 6$
$n = 7$

d $7, 14, 28, \ldots, 224$

$a = 7, r = 2$
$a_n = ar^{n-1}$
$224 = 7(2^{n-1})$
$32 = 2^{n-1}$
$2^5 = 2^{n-1}$
$n - 1 = 5$
$n = 6$

e $2, 14, 98, \ldots, 4802$

$a = 2, r = 7$
$a_n = ar^{n-1}$
$4802 = 2(7^{n-1})$
$2401 = 7^{n-1}$
$7^4 = 7^{n-1}$
$n - 1 = 4$
$n = 5$

f $1/25, 1/5, 1, \ldots, 625$

$a = \frac{1}{25}, r = 5$
$a_n = ar^{n-1}$
$625 = \left(\frac{1}{25}\right)(5^{n-1})$
$15625 = 5^{n-1}$
$5^6 = 5^{n-1}$
$n - 1 = 6$
$n = 7$

10

a The first term of a GP is $a = 25$ and the fourth term is $a_4 = 200$.
Use the formula $a_n = ar^{n-1}$ to find the common ratio r, then write down the first five terms.

$a_4 = ar^3$
$200 = 25r^3$
$r^3 = 8$
$r = 2$
First five terms: 25, 50, 100, 200, 400

b Find the common ratio r of a GP for which:

i $a = 3 \quad a_6 = 96$

2

ii $a = 1000 \quad a_7 = 0.001$

0.1 or –0.1

iii $a = 32 \quad a_6 = -243$

$-\frac{3}{2}$

iv $a = 5 \quad a_7 = 40$

$\pm\sqrt{2}$

11 The nth term of a geometric sequence is $a_n = 25 \times 2^n$.

a Write out the first six terms and hence find the values of a and r.

50, 100, 200, 400, 800, 1600,

$a = 50, r = 2$

b Which term of the sequence is 6400?

$25 \times 2^n = 6400$

$2^n = 256$

$2^n = 2^8$

$n = 8$

c Prove that $a_9 \times a_{11} = 25 \times a_{20}$.

$a_9 \times a_{11} = (25 \times 2^9)(25 \times 2^{11})$

$= 625 \times 2^{20}$

$= 25 \times 25 \times 2^{20}$

$= 25 \times (25 \times 2^{20})$

$= 25 \times a_{20}$ as required

d Write out the terms between 1000 and 100000. How many of them are there?

$a_6 = 1600$, $a_7 = 3200$, $a_8 = 6400$, $a_9 = 12800$, $a_{10} = 25600$, $a_{11} = 51200$

∴ The six terms a_6 , ..., a_{11} lie between 1000 and 100000

e Verify by calculations that $a_{11} = 51200$ is the last term less than $100\,000$ and that $a_{12} = 102400$ is the first term greater than $100\,000$.

$a_{11} = 25 \times 2^{11} = 25 \times 2048 = 51200$

$a_{12} = 25 \times 2^{12} = 25 \times 4096 = 102400$

12 A piece of paper 0.1 mm thick is folded successively 100 times. How thick is it now?

GP with $a = 0.1, r = 2, n = 101$ terms

$a_{101} = ar^{100}$

$= 0.1 \times 2^{100}$

$= \frac{2^{100}}{10}$ mm

∴ Thickness ≈ 1.27×10^{23} km

This is around 5200 times the distance from Earth to the nearest galaxy.

13 Write down the first few terms of the GP generated by substituting the positive integers into the exponential function $f(x) = \frac{4}{25} \times 5^x$. Then write down a formula for the nth term.

f(1) = (4/25) × 5 = 4/5

f(2) = (4/25) × 25 = 4

f(3) = (4/25) × 125 = 20

f(4) = (4/25) × 625 = 100

Geometric sequences are equivalent to exponential functions (but the domain is restricted to positive whole numbers)

$a_n = \left(\frac{4}{25}\right) \times 5^n$

14 A principal $\$P$ is invested at 7% per annum compound interest.
Let A_n be the total amount at the end of n years.

a Write down A_1, A_2 and A_3.

$A_1 = P \times 1.07$

$A_2 = P \times (1.07)^2$

$A_3 = P \times (1.07)^3$

b Find the nth term A_n.

$A_n = ar^{n-1}$

$= (1.07P)(1.07^{n-1})$

$= P \times (1.07)^n$

c Use trial-and-error on the calculator to find how many full years it will take for the amount to double, and how many years it will take for it to exceed ten times the original principal.

Let $A_n = 10P$

$P \times (1.07)^n > 10P$

$1.07^n > 10$

By trial and error, $n = 35$

d Use logarithms to confirm your answer.

$1.07^n > 10$

$n > \log_{1.07} 10$

$n > 34.03$

$\therefore n = 35$

15 A car originally costs \$20000, then at the end of every year, it is worth only 80% of what it was worth a year before. Let W_n be its worth at the end of n years.

a Write down expressions for W_1, W_2 and W_3, and find a formula for W_n.

$W_1 = 20000 \times 0.8$

$W_2 = 20000 \times (0.8)^2$

$W_3 = 20000 \times (0.8)^3$

GP with $a = 20000 \times 0.8$ and $r = 0.8$

$W_n = ar^{n-1} = (20000 \times 0.8)(0.8^{n-1})$

$= 20000(0.8)^n$

b Use trial-and-error on the calculator to find how many complete years it takes for the value to fall below \$2000.

$20000(0.8)^n < 2000$

$(0.8)^n < 0.1$

By trial and error, $n = 11$

c Use logarithms to solve part (b). When you log both sides, if the base of the logarithm is between 0 and 1, you must reverse the inequality.

$(0.8)^n < 0.1$

$n > \log_{0.8} 0.1$

$n > 10.31\ldots$

$\therefore n = 11$

16 Find the nth term of each GP in fully simplified form.

a $\sqrt{6}, 2\sqrt{3}, 2\sqrt{6}, \ldots$

$a = \sqrt{6}, r = \sqrt{2}$

$a_n = ar^{n-1}$

$= \sqrt{6} \times (\sqrt{2})^{n-1}$

$= \sqrt{6} \times \sqrt{2^{n-1}}$

$= \sqrt{6 \times 2^{n-1}}$

$= \sqrt{3} \times \sqrt{2 \times 2^{n-1}}$

$= \sqrt{3} \times (\sqrt{2})^n$

b $ax, a^2x^3, a^3x^5, \ldots$

$a = ax, r = ax^2$

$a_n = ar^{n-1}$

$= ax \times (ax^2)^{n-1}$

$= a^n x^{2n-1}$

c $-\frac{x}{y}, -1, -\frac{x}{y^2}, \ldots$

$a = -\frac{x}{y}, r = \frac{y}{x}$

$a_n = ar^{n-1}$

$= \left(-\frac{x}{y}\right) \times \left(\frac{y}{x}\right)^{n-1}$

$= -x^{(2-n)} y^{n-2}$

17 The following are GPs.

a Find a formula for a_n in $2x, 2x^2, 2x^3, \ldots$ Then find x if $a_6 = 2$.

$a = 2x, r = x$

$a_n = ar^{n-1} = 2x \times x^{n-1} = 2x^n$

If $a_6 = 2$:

$2x^6 = 2$

$x^6 = 1$

$x = 1$ or -1

b Find a formula for a_n in $x^4, x^2, 1, \ldots$ Then find x if $a_6 = 3^6$.

$a = x^4, r = \frac{1}{x^2}$

$a_n = ar^{n-1}$

$= x^4 \times \left(\frac{1}{x^2}\right)^{n-1}$

$= x^{6-2n}$

If $a_6 = 3^6$:

$x^{6-12} = 3^6$

$x^{-6} = 3^6$

$\frac{1}{x^6} = 729$

$x^6 = \frac{1}{729}$

$x = \frac{1}{3}$ or $-\frac{1}{3}$

c Find a formula for a_n in $2^{-16}x, 2^{-12}x, 2^{-8}x, \ldots$ Then find x if $a_6 = 96$.

$a = 2^{-16}x, r = 2^4$

$a_n = ar^{n-1}$

$= 2^{-16}x \times (2^4)^{n-1}$

$= 2^{-16}x \times 2^{4n-4}$

$= 2^{(4n-20)}x$

If $a_6 = 96$:

$2^{24-20}x = 96$

$2^4 x = 96$

$16x = 96$

$x = 6$

Partial Sum of a Geometric Series

Investigation Partial sum of a Geometric Series.

Consider the partial sum of a geometric sequence below:

$$S_n = a + ar + ar^2 + ar^3 + \cdots ar^{n-1}$$

A quick way of finding the partial sum is to multiply both sides by r.

$$rS_n = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

Subtract the two equations above:

$$rS_n - S_n = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

Then rearrange for S_n

$$S_n = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

By multiplying the numerator and denominator, we can get an alternate formula:

$$S_n = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

ⓘ The Partial Sum of a Geometric Series.

$S_n = \frac{a(r^n - 1)}{r - 1}$	$S_n = \frac{a(1 - r^n)}{1 - r}$

⚠ Usually n is not given, you need to find it using $a_n = ar^{n-1}$

Example Finding the partial sum of an arithmetic series.

Consider the geometric series: $2 - 6 + 18 - \cdots$
Find S_6

$a = 2, r = -3$

$$S_6 = \frac{2((-3)^6 - 1)}{(-3) - 1}$$
$$= -364$$

Practice

Consider the geometric series: $60 + 20 + \frac{20}{3} + \cdots$
Find S_{10}

89.99847584…

Consider the geometric series: $3 + 12 + 48 + \cdots + 786\,432$
Find the number of terms in the partial sum and hence evaluate the partial sum of the series.

1 048 575

Example Further problems involving sums of geometric sequences.

The sum of the first 4 terms of a geometric sequence with common ratio 3 is 200.
Find the first term.

$$S_4 = 200$$

$$\frac{a(3^4 - 1)}{3 - 1} = 200$$

$$\frac{80a}{2} = 200$$

$$40a = 200$$

$$a = 5$$

Practice

Find the value of n for which the partial sum S_n of $1 + 4 + 16 + \cdots$ is equal to 21 845.

8

How many terms of the series $2 + 6 + 18 + \cdots$ are needed to give a sum of one billion?

19

1

a Use the formula $S_7 = \frac{a(r^7-1)}{r-1}$ to find $1 + 3 + 3^2 + 3^3 + 3^4 + 3^5 + 3^6$.

$a = 1, r = 3, n = 7$

$s_7 = \frac{1(3^7 - 1)}{3 - 1}$

$= 1093$

b Use the formula $s_7 = \frac{a(1-r^7)}{1-r}$ to find $1 - 3 + 3^2 - 3^3 + 3^4 - 3^5 + 3^6$.

$a = 1, r = -3, n = 7$

$S^7 = \frac{1(1 - (-3)^7)}{1 - (-3)}$

$= 547$

2 Find these sums using $S_n = \frac{a(r^n - 1)}{r-1}$ when $r > 1$, or $S_n = \frac{a(1-r^n)}{1-r}$ when $r < 1$.

a 1 + 2 + 4 + 8 + ⋯ (10 terms)

$a = 1, r = 2, n = 10$

$S_{10} = \frac{1(2^{10} - 1)}{2 - 1}$

$= 1023$

b 2 + 6 + 18 + ⋯ (5 terms)

$a = 2, r = 3, n = 5$

$S_5 = \frac{2(3^5 - 1)}{3 - 1}$

$= 242$

c −1 − 10 − 100 − ⋯ (5 terms)

$a = -1, r = 10, n = 5$

$S_5 = -\frac{1(10^5 - 1)}{10 - 1}$

$= -11111$

d −1 − 5 − 25 − ⋯ (5 terms)

$a = -1, r = 5, n = 5$

$S_5 = -\frac{1(5^5 - 1)}{5 - 1}$

$= -781$

e 1 − 2 + 4 − 8 + ⋯ (10 terms)

$a = 1, r = -2, n = 10$

$s_{10} = \frac{1(1 - (-2)^{10})}{1 - (-2)}$

$= -341$

f 2 − 6 + 18 − ⋯ (5 terms)

$a = 2, r = -3, n = 5$

$S_5 = \frac{2(1 - (-3)^5)}{1 - (-3)}$

$= 122$

g −1 + 10 − 100 + ⋯ (5 terms)

$a = -1, r = -10, n = 5$

$S_5 = -\frac{1(1 - (-10)^5)}{1 - (-10)}$

$= -9091$

h −1 + 5 − 25 + ⋯ (5 terms)

$a = -1, r = -5, n = 5$

$S_5 = -\frac{1(1 - (-5)^5)}{1 - (-5)}$

$= -521$

3 Find these sums.

a 8 + 4 + 2 + ⋯ (10 terms)

$a = 8, r = \frac{1}{2}, n = 10$

$S_{10} = \frac{8\left(1 - \left(\frac{1}{2}\right)^{10}\right)}{1 - \frac{1}{2}}$

$= \frac{1023}{64}$

b 9 + 3 + 1 + ⋯ (6 terms)

$a = 9, r = \frac{1}{3}, n = 6$

$S_6 = \frac{9\left(1 - \left(\frac{1}{3}\right)^{6}\right)}{1 - \frac{1}{3}}$

$= \frac{364}{27}$

c 45 + 15 + 5 + ⋯ (5 terms)

$a = 45, r = \frac{1}{3}, n = 5$

$S_5 = \frac{45\left(1 - \left(\frac{1}{3}\right)^{5}\right)}{\frac{2}{3}}$

$= \frac{605}{9}$

d $\frac{2}{3} + 1 + \frac{3}{2} + \frac{9}{4} + \frac{27}{8}$

$a = \frac{2}{3}, r = \frac{3}{2}, n = 5$

$S_5 = \frac{\left(\frac{2}{3}\right)\left(\left(\frac{3}{2}\right)^{5} - 1\right)}{\frac{3}{2} - 1}$

$= \frac{211}{24}$

e 8 − 4 + 2 − ⋯ (10 terms)

$a = 8, r = -\frac{1}{2}, n = 10$

$S_{10} = \frac{8\left(1 - \left(-\frac{1}{2}\right)^{10}\right)}{1 - \left(-\frac{1}{2}\right)}$

$= \frac{341}{64}$

f 9 − 3 + 1 − ⋯ (6 terms)

$a = 9, r = -\frac{1}{3}, n = 6$

$S_6 = \frac{9\left(1 - \left(-\frac{1}{3}\right)^{6}\right)}{1 - \left(-\frac{1}{3}\right)}$

$= \frac{182}{27}$

g −45 + 15 − 5 + ⋯ (5 terms)

$a = -45, r = -\frac{1}{3}, n = 5$

$S_5 = -\frac{45\left(1 - \left(-\frac{1}{3}\right)^{5}\right)}{1 - \left(-\frac{1}{3}\right)}$

$= -\frac{305}{9}$

h $\frac{2}{3} - 1 + \frac{3}{2} - \frac{9}{4} + \frac{27}{8}$

$a = \frac{2}{3}, r = -\frac{3}{2}, n = 5$

$S_5 = \frac{\left(\frac{2}{3}\right)\left(1 - \left(-\frac{3}{2}\right)^{5}\right)}{1 - \left(-\frac{3}{2}\right)}$

$= \frac{55}{24}$

4 The King takes a chessboard of 64 squares, and places 1 grain of wheat on the first square, 2 grains on the next square, 4 grains on the next square, and so on.
How many grains are on:

a the last square? Answer using scientific notation to 2 significant figures.

$a = 1, r = 2, n = 64$

$a_{64} = ar^{63}$

$= 1(2^{63})$

$= 2^{63}$

$\approx 9.2 \times 10^{18}$

b The whole chessboard? Answer in index form.

$S^{64} = \frac{a(r^{64} - 1)}{r - 1}$

$= \frac{1(2^{64} - 1)}{2 - 1}$

$= 2^{64} - 1$

5 Find these sums. First write out some terms and identify a and r.

a

$$\sum_{n=1}^{7} 3 \times 2^n$$

$6 + 12 + 24 \ldots$
$a = 6, r = 2, n = 7$
$S_7 = \frac{6(2^7 - 1)}{2 - 1}$
$= 762$

b

$$\sum_{n=3}^{8} 3^{n-1}$$

$9 + 27 + 81 \ldots$
$a = 9, r = 3, n = 6$
$S_6 = \frac{9(3^6 - 1)}{3 - 1}$
$= 3276$

c

$$\sum_{n=1}^{8} 3 \times 2^{3-n}$$

$12 + 6 + 3 \ldots$
$a = 12, r = \frac{1}{2}, n = 8$
$S^8 = \frac{12\left(1 - \left(\frac{1}{2}\right)^8\right)}{\frac{1}{2}}$
$= \frac{765}{32}$

6

a The first term of a GP is $\frac{1}{8}$ and the fifth term is 162.
Find the first five terms of the GP, then find their sum. There are two possibilities.

$a = \frac{1}{8}, a_5 = 162$
$a_5 = ar^4$
$162 = \left(\frac{1}{8}\right)r^4$
$r^4 = 1296$
$r = \pm 6$
Taking $r = 6$:
$S_5 = \frac{\left(\frac{1}{8}\right)(6^5 - 1)}{6 - 1}$
$= \frac{1555}{8}$
Taking $r = -6$:
$S_5 = \frac{\left(\frac{1}{8}\right)(1 - (-6)^5)}{1 - (-6)}$
$= \frac{1111}{8}$
$\therefore \frac{1555}{8}$ or $\frac{1111}{8}$

b The first term of a GP is $-\frac{3}{4}$ and the fourth term is 6. Find the sum of the first six terms.

$a = -\frac{3}{4}, a_4 = 6$
$a_4 = ar^3$
$6 = \left(-\frac{3}{4}\right)r^3$
$r^3 = -8$
$r = -2$
$S_6 = \frac{a(1 - r^6)}{1 - r}$
$= \frac{\left(-\frac{3}{4}\right)(1 - (-2)^6)}{1 - (-2)}$
$= 15.75$

c The second term of GP is 0.08 and the third term is 0.4. Find the sum to eight terms.

$$a_2 = 0.08, a_3 = 0.4$$

$$r = \frac{a_3}{a_2} = 5$$

$$\text{also, } r = \frac{a_2}{a}$$

$$\therefore a = \frac{a_2}{r} = \frac{0.08}{5} = 0.016$$

$$S_8 = \frac{a(r^8 - 1)}{r - 1}$$

$$= \frac{0.016(5^8 - 1)}{5 - 1}$$

$$= 1562.496$$

d The ratio of a GP is $r = 2$ and the sum to eight terms is 1785. Find the first term.

$$r = 2, S_8 = 1785$$

$$S_8 = \frac{a(r^8 - 1)}{r - 1}$$

$$1785 = \frac{a(2^8 - 1)}{2 - 1}$$

$$1785 = 255a$$

$$a = 7$$

e A GP has ratio $r = -\frac{1}{2}$ and the sum to eight terms is 425. Find the first term.

$$r = -\frac{1}{2}, S_8 = 425$$

$$S_8 = \frac{a(1 - r^8)}{1 - r}$$

$$425 = \frac{a\left(1 - \left(-\frac{1}{2}\right)^8\right)}{1 - \left(-\frac{1}{2}\right)}$$

$$425 = \frac{a\left(1 - \frac{1}{256}\right)}{\left(\frac{3}{2}\right)}$$

$$\frac{1275}{2} = \frac{255}{256}a$$

$$a = 640$$

7

a Each year when the sunflower paddock is weeded, only half the previous weight of weed is dug out. In the first year, 6 tonnes of weed is dug out.

i How much is dug out in the 10th year?

$$a = 6, r = \frac{1}{2}$$

$$a_{10} = ar^9$$

$$= 6\left(\frac{1}{2}\right)^9$$

$$= \frac{3}{256}$$

$$\approx 0.01172 \text{ tonnes}$$

ii What is the total dug out over 10 years (correct to four significant figures)?

$$S_{10} = \frac{a(1 - r^{10})}{1 - r} = \frac{6\left(1 - \left(\frac{1}{2}\right)^{10}\right)}{\frac{1}{2}}$$

$$\approx 11.99 \text{ tonnes}$$

8 Every two hours, half of a particular medical isotope decays. If there was originally 20 grams, how much remains after a day (correct to two significant figures)?

$$a = 20, r = \frac{1}{2}, 1 \text{ day} = 12 \text{ two hour periods}$$

$$a_{12} = ar^{11}$$

$$= 20\left(\frac{1}{2}\right)^{11}$$

$$= \frac{20}{2048}$$

$$\approx 0.009766 \text{ g}$$

$$\approx 4.9 \times 10^{-3} \text{ g}$$

9 The price of Victoria shoes is increasing over a 10-year period by 10% per annum, so that the price in each of those 10 years is $P, 1.1\,P, (1.1)^2P$, ... I buy one pair of these shoes each year.

a Find an expression for the total paid over 10 years (correct to the nearest cent).

$$a = P, r = 1.1, n = 10$$

$$S_{10} = \frac{a(r^{10} - 1)}{r - 1}$$

$$= \frac{P((1.1)^{10} - 1)}{1.1 - 1}$$

$$= \frac{P((1.1)^{10} - 1)}{0.1}$$

$$= 10P((1.1)^{10} - 1)$$

b Hence find the initial price P if the total paid is \$900.

$$900 = 10P((1.1)^{10} - 1)$$

$$P = \frac{900}{10(1.1^{10} - 1)}$$

$$P \approx 56.47$$

10 The number of people attending the yearly Abletown Show is rising by 5% per annum, and the number attending the yearly Bush Creek Show is falling by 5% per annum. In the first year under consideration, 5000 people attended both shows.

a Find the total number attending each show during the first six years.

Abletown: GP where $a = 5000, r = 1.05, n = 6$

$$S_6 = \frac{5000((1.05)^6 - 1)}{1.05 - 1}$$

= 34010

Bush Creek: GP where $a = 5000, r = 0.95, n = 6$

$$S_6 = \frac{5000(1 - (0.95)^6)}{1 - 0.95}$$

= 26491

b What is the ratio (correct to three significant figures) of the total number attending the Abletown Show over these 25 years to the total attending the Bush Creek Show?

$$\text{Abletown } S_{25} = \frac{5000((1.05)^{25} - 1)}{0.05}$$

$$\text{Bush Creek } S_{25} = \frac{5000(1 - (0.95)^{25})}{0.05}$$

$$\text{Ratio} = \frac{(1.05)^{25} - 1}{1 - (0.95)^{25}}$$

$$\approx 3.30$$

11

a Show that the sum S_n of the first n terms of $7 + 14 + 28 + \cdots$ is $S_n = 7(2^n - 1)$.

$a = 7, r = 2$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$= \frac{7(2^n - 1)}{2 - 1}$

$= 7(2^n - 1)$ as required

b For what value of n is S_n equal to 1785?

$7(2^n - 1) = 1785$

$2^n - 1 = 255$

$2^n = 256$

$2^n = 2^8$

$n = 8$

c Show that $a_n = 7 \times 2^{n-1}$, and find how many terms are less than 70000.

$a_n = ar^{n-1} = 7 \times 2^{n-1}$

$a_n < 70000$:

$7 \times 2^{n-1} < 70000$

$2^{n-1} < 10000$

$n - 1 < \log_2 10000$

$n < \log_2 10000 + 1$

$n < 14.29\ldots$

∴ 14 terms are less than 70000

d Prove that the sum S_n of the first n terms is always 7 less than the $(n + 1)^{\text{th}}$ term.

Want to show that $a_{n+1} - 7 = S_n$

$S_n = 7(2^n - 1)$

$a_{n+1} - 7$

$= (7 \times 2^n) - 7$

$= 7(2^n - 1)$

∴ $S_n = a_{n+1} - 7$ as required

12 The powers of 3 that are greater than 1 form a GP 3, 9, 27, ...

a Find using logarithms how many powers of 3 there are between 2 and 10^{20}.

$2 < 3^n < 10^{20}$

$\log_{10} 2 < n \log_{10} 3 < 20$

$0.631\ldots < n < 41.92\ldots$

∴ $n = 1, 2, 3, \ldots, 41$

∴ 41 powers of 3

b Show that $S_n = \frac{3}{2}(3^n - 1)$, and find the smallest value of n for which $S_n > 10^{20}$.

$a = 3, r = 3$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$= \frac{3(3^n - 1)}{3 - 1}$

$= \frac{3(3^n - 1)}{2}$

$= \frac{3}{2(3^n - 1)}$ as required

Need $S_n > 10^{20}$:

$\frac{3}{2}(3^n - 1) > 10^{20}$

$3^n - 1 > \left(\frac{2}{3}\right) \times 10^{20}$

$3^n > \left(\frac{2}{3}\right) \times 10^{20} + 1$

$n > \log_3\left(\left(\frac{2}{3}\right) \times 10^{20} + 1\right)$

$n > 41.54\ldots$

$\therefore n = 42$ terms

13 Find a formula for S_n, and hence find n for the given value of S_n.

a 5 + 10 + 20 + ⋯ where $S_n = 315$

$a = 5, r = 2$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$= \frac{5(2^n - 1)}{2 - 1}$

$= 5(2^n - 1)$

$315 = 5(2^n - 1)$

$63 = 2^n - 1$

$2^n = 64$

$2^n = 2^6$

$n = 6$

b 5 − 10 + 20 − ⋯ where $S_n = -425$

$a = 5, r = -2$

$S_n = \frac{a(1 - r^n)}{1 - r}$

$= \frac{5(1 - (-2)^n)}{1 - (-2)}$

$= \frac{5(1 - (-2)^n)}{3}$

$-425 = \frac{5(1 - (-2)^n)}{3}$

$-1275 = 5(1 - (-2)^n)$

$-255 = 1 - (-2)^n$

$(-2)^n = 256$

$n = 8$

c 18 + 6 + 2 + ⋯ where $S_n = 26\frac{8}{9}$

$a = 18, r = \frac{1}{3}$

$S_n = \frac{a(1 - r^n)}{1 - r}$

$= \frac{18\left(1 - \left(\frac{1}{3}\right)^n\right)}{\frac{2}{3}}$

$= 27\left(1 - \left(\frac{1}{3}\right)^n\right)$

$26\frac{8}{9} = 27\left(1 - \left(\frac{1}{3}\right)^n\right)$

$\frac{242}{9} = 27\left(1 - \left(\frac{1}{3}\right)^n\right)$

$\frac{242}{243} = 1 - \left(\frac{1}{3}\right)^n$

$\left(\frac{1}{3}\right)^n = \frac{1}{243}$

$\left(\frac{1}{3}\right)^n = \left(\frac{1}{3}\right)^5$

$n = 5$

d 48 − 24 + 12 − ⋯ where $S_n = 32\frac{1}{4}$

$a = 48, r = -\frac{1}{2}$

$S_n = \frac{a(1 - r^n)}{1 - r}$

$= \frac{48\left(1 - \left(-\frac{1}{2}\right)^n\right)}{\frac{3}{2}}$

$= 32\left(1 - \left(-\frac{1}{2}\right)^n\right)$

$32\frac{1}{4} = 32\left(1 - \left(-\frac{1}{2}\right)^n\right)$

$\frac{129}{4} = 32\left(1 - \left(-\frac{1}{2}\right)^n\right)$

$\frac{129}{128} = 1 - \left(-\frac{1}{2}\right)^n$

$\left(-\frac{1}{2}\right)^n = -\frac{1}{128}$

$n = 7$

Limiting Sum of a Geometric Series

Investigation Limiting sum of a geometric series.

Consider the infinite geometric series below:

$$1 + 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + 512 + \cdots$$

Try adding the first 10 terms of this series. What do you notice about the partial sums?

Consider this infinite geometric series:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \frac{1}{128} + \frac{1}{512} + \cdots$$

Try adding the first 10 terms of this series using a calculator. What do you notice?

For some geometric series, there is a **limiting sum** as $n \to \infty$.

https://www.desmos.com/calculator/nygpeq1xgg

Investigate what happens to the partial sum of a geometric series when:

$r > 1$	$0 < r < 1$	$-1 < r < 0$	$r < -1$

A geometric series has a **limiting sum** when ……

Let's confirm this by analysing the limit of the sum:

$$s_\infty = \lim_{n \to \infty} \frac{a(1 - r^n)}{1 - r}$$

When $|r| > 1$,

$S_\infty =$

When $|r| < 1$,

$S_\infty =$

ⓘ The Limiting Sum of Geometric Series

When $|r| < 1$,

the infinite sum of a geometric series **converges** to a value, called the **limiting sum**.

$$S_\infty = \frac{a}{1-r}$$

ⓘ **Identify** whether a geometric series has a limiting sum

✗ 6 + 12 + 24 + 48 +…

✓ 6 + 3 + 1.5 + 0.75…

✓ 6 – 3 + 1.5 – 0.75…

✗ 6 + 6 + 6 + 6 + …

Determine whether the following series will converge to a limiting sum.
For those that do, find the limiting sum.

a 10 + 5 + 2.5 + 1.25…	**b** 10000 + 1000 + 100 + ..	**c** 14 + 11 + 8 + 5 + 2…
d $40 + \frac{80}{3} + \frac{160}{9} + \frac{320}{27} \ldots$	**e** $\sum_{n=1}^{\infty} 3\left(\frac{4}{5}\right)^{n-1}$	**f** $\sum_{n=1}^{\infty} 3\left(\frac{5}{4}\right)^{n-1}$

Example Further problems involving limiting sums of geometric series.

Find the common ratio of a geometric sequence whose first term is 10 and limiting sum is 40.

$$s_\infty = \frac{a}{1-r}$$
$$40 = \frac{10}{1-r}$$
$$1-r = \frac{10}{40}$$
$$-r = -\frac{3}{4}$$
$$r = \frac{3}{4}$$

Practice

Find the common ratio of a geometric sequence whose first term is 15 and limiting sum is 25.

$\frac{2}{5}$

Find the first term of a geometric sequence whose common ratio is 0.5 and limiting sum is 40.

20

1

a Complete the table of values for the GP with $a = 18$ and $r = \frac{1}{3}$.

n	1	2	3	4	5	6
a_n	18	6	2	$\frac{2}{3}$	$\frac{2}{9}$	$\frac{2}{27}$
S_n	18	24	26	$26\frac{2}{3}$	$26\frac{8}{9}$	$26\frac{26}{27}$

b Find the limiting sum using $S_\infty = \frac{a}{1-r}$.

$S_\infty = \frac{a}{1-r} = \frac{18}{1-\frac{1}{3}} = 27$

c Find the difference $S_\infty - S_6$.

$S_\infty - S_6 = 27 - 26(26/27)$

$= 1\frac{26}{27}$

2

a Complete the table of values for the GP with $a = 24$ and $r = -\frac{1}{2}$.

n	1	2	3	4	5	6
a_n	24	-12	6	-3	$\frac{3}{2}$	$-\frac{3}{4}$
S_n	24	12	18	15	$16\frac{1}{2}$	$15\frac{3}{4}$

b Find the limiting sum using $S_\infty = \frac{a}{1-r}$.

$S_\infty = \frac{a}{1-r} = \frac{24}{1-\left(-\frac{1}{2}\right)} = 16$

3 Identify the first term a and ratio r of each GP and hence find S_∞.

a $8 + 4 + 2 + \cdots$

$a = 8, r = \frac{1}{2}$

$s_\infty = 16$

b $-4 - 2 - 1 - \cdots$

$a = -4, r = \frac{1}{2}$

$s_\infty = -8$

c $1 - \frac{1}{3} + \frac{1}{9} - \cdots$

$a = 1, r = -\frac{1}{3}$

$s_\infty = \frac{3}{4}$

d $36 - 12 + 4 - \cdots$

$a = 36, r = -\frac{1}{3}$

$s_\infty = 27$

e $60 - 30 + 15 - \cdots$

$a = 60, r = -\frac{1}{2}$

$s_\infty = 40$

f $60 - 12 + 2\left(\frac{2}{5}\right) - \cdots$

$a = 60, r = -\frac{1}{5}$

$s_\infty = 50$

4 Find each ratio r to test whether there is a limiting sum. Find the limiting sum if it exists.

a $1 - \frac{1}{2} + \frac{1}{4} - \cdots$

$|r| = \left|-\frac{1}{2}\right| < 1$,

$S_\infty = \frac{2}{3}$

b $4 - 6 + 9 - \cdots$

$|r| = \left|\frac{3}{2}\right| > 1$, so, no limiting sum

c $12 + 4 + \frac{4}{3} + \cdots$

$S_\infty = 18$

d $1000 + 100 + 10 + \cdots$

$S_\infty = \frac{10000}{9}$

e $-2 + \frac{2}{5} - \frac{2}{25} + \cdots$

$S_\infty = -\frac{5}{3}$

f $-\frac{2}{3} - \frac{2}{15} - \frac{2}{75} - \cdots$

$S_\infty = -\frac{5}{6}$

5 Bevin dropped a bouncy ball from a height of 8 metres. It bounced continually, each successive height being half of the previous height.

a Show that the first distance travelled down-and-up is 12 metres, and explain why the successive down-and-up distances form a GP with $r = \frac{1}{2}$.

Ball drops 8 m, bounces to 4 m

First down-and-up distance = 12 m

Each bounce is $\frac{1}{2}$ of previous height, so distances form GP with $a = 12, r = \frac{1}{2}$

b Through what distance did the ball 'eventually' travel?

$a = 12, r = \frac{1}{2}$

$S_\infty = 24$

6 When a council began offering free mailboxes to its 10000 home owners, 20% installed them in the first month. The number installing them in the second month was only 20% of those in the first month, and so on.

a Show that the homeowners installing them each month form a GP.

2000, 400, 80…

$a = 2000, r = 0.2$

b How many homeowners will 'eventually' install them? ('Eventually' means take S_∞.)

$S_\infty = 2500$

7 A factory has been advertising its unbreakable widgets every month.
The first advertisement brought in 1000 sales, but every successive advertisement is only bringing in 90% of the previous month's sales.

a How many widget sales will the advertisements 'eventually' bring in?

$a = 1000, r = 0.9$

$S_\infty = 10000$

b About how many eventual sales were not brought in by the first 10 advertisements?

$S_\infty - S_{10} = 3487$

8 Find the limiting sums if they exist, rationalising denominators where necessary.

a $1 + (1.01) + (1.01)^2 +$

$r = 1.01 > 1$
no limiting sum

b $1 + (1.01)^{-1} + (1.01)^{-2} +$

$S_\infty = 101$

c $16\sqrt{5} + 4\sqrt{5} + \sqrt{5} + \cdots$

$S_\infty = \frac{54\sqrt{5}}{3}$

d $7 + \sqrt{7} + 1 + \cdots$

$S_\infty = \frac{7(7 + \sqrt{7})}{6}$

e $4 - 2\sqrt{2} + 2 - \cdots$

$S_\infty = 4(2 - \sqrt{2})$

f $5 - 2\sqrt{5} + 4 - \cdots$

$S_\infty = 5(5 - 2\sqrt{5})$

g $9 + 3\sqrt{10} + 10 + \cdots$

$r = \frac{\sqrt{10}}{3} > 1$
No limiting sum

h $1 + (1 - \sqrt{3}) + (1 - \sqrt{3})^2 + \cdots$

$S_\infty = \frac{\sqrt{3}}{3}$

9 Expand each series for a few terms. Then write down a and r, and find the limiting sum.

a $\sum_{n=1}^{\infty} \left(\frac{1}{5}\right)^n$

$\frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \cdots$
$a = \frac{1}{5}, r = \frac{1}{5}$
$S_\infty = \frac{1}{4}$

b $\sum_{n=1}^{\infty} 7\left(\frac{1}{2}\right)^n$

$\frac{7}{2} + \frac{7}{4} + \frac{7}{8} + \cdots$
$a = \frac{7}{2}, r = \frac{1}{2}$
$S_\infty = 7$

c $\sum_{n=1}^{\infty} 40\left(-\frac{3}{5}\right)^n$

$-24 + \frac{72}{5} - \frac{216}{25} + \cdots$
$a = -24, r = -\frac{3}{5}$
$S_\infty = -15$

10 Find, in terms of x, an expression for the limiting sum of the series on the LHS of each equation. Then solve the equation to find x.

a $5 + 5x + 5x^2 + \cdots = 10$

$S_\infty = \frac{a}{1 - r}$
$= \frac{5}{1 - x}$

$\frac{5}{1 - x} = 10$
$5 = 10 - 10x$
$10x = 5$
$x = \frac{1}{2}$

b $5 - 5x + 5x^2 - \cdots = 15$

$S_\infty = \frac{a}{1 - r}$
$= \frac{5}{1 - (-x)}$
$= \frac{5}{1 + x}$

$\frac{5}{1 + x} = 15$
$x = -\frac{2}{3}$

c $x + \frac{x}{3} + \frac{x}{9} + \cdots = 2$

$S_\infty = \frac{x}{1 - \frac{1}{3}}$
$= \frac{3x}{2}$

$\frac{3x}{2} = 2$
$x = \frac{4}{3}$

d $x - \frac{x}{3} + \frac{x}{9} - \cdots = 2$

$S_\infty = \frac{x}{1 - \left(-\frac{1}{3}\right)}$
$= \frac{3x}{4}$

$\frac{3x}{4} = 2$
$3x = 8$
$x = \frac{8}{3}$

11 Find the condition for each GP to have a limiting sum, then find that limiting sum.

a $7 + 7x + 7x^2 + \cdots$

$r = x$
For limiting sum: $|r| < 1$
$\therefore -1 < x < 1$

$S_\infty = \frac{a}{1-r}$
$= \frac{7}{1-x}$

b $2x + 6x^2 + 18x^3 + \cdots$

$r = 3x$
For limiting sum: $|3x| < 1$
$\therefore -\frac{1}{3} < x < \frac{1}{3}$

$S_\infty = \frac{a}{1-r}$
$= \frac{2x}{1-3x}$

c $1 + (x-1) + (x-1)^2 + \cdots$

$r = x - 1$
For limiting sum: $|x - 1| < 1$
$-1 < x - 1 < 1$
$0 < x < 2$

$S_\infty = \frac{a}{1-r}$
$= \frac{1}{1-(x-1)}$
$= \frac{1}{2-x}$

d $1 + (1+x) + (1+x)^2 + \cdots$

$r = 1 + x$
For limiting sum: $|1 + x| < 1$
$-1 < 1 + x < 1$
$-2 < x < 0$

$S_\infty = \frac{a}{1-r}$
$= \frac{1}{1-(1+x)}$
$= -\frac{1}{x}$

12 Find the condition for each GP to have a limiting sum, then find that limiting sum.

a $1 + (x^2-1) + (x^2-1)^2 + \cdots$

$r = x^2 - 1$
For limiting sum: $|x^2 - 1| < 1$
$-1 < x^2 - 1 < 1$
$0 < x^2 < 2$
$-\sqrt{2} < x < \sqrt{2}$ and $x \neq 0$

$S_\infty = \frac{a}{1-r}$
$= \frac{1}{1-(x^2-1)}$
$= \frac{1}{2-x^2}$

b $1 + \frac{1}{1+x^2} + \frac{1}{(1+x^2)^2} + \cdots$

$r = \frac{1}{1+x^2}$
For limiting sum: $\left|\frac{1}{1+x^2}\right| < 1$ (always true for $x \neq 0$)

$x \neq 0$
$S_\infty = \frac{a}{1-r}$
$= \frac{1}{1-\frac{1}{1+x^2}}$
$= \frac{1}{\frac{1+x^2-1}{1+x^2}}$
$= \frac{1}{\frac{x^2}{1+x^2}}$
$= \frac{1+x^2}{x^2}$

13 The series 4 + 12 + 36 + ⋯ has no limiting sum because r > 1.
If we substitute into the formula for the limiting sum, we get:

$$S_\infty = \frac{4}{1-3} = -2.$$

Can any meaning be given to this calculation and its result?
Use DESMOS to investigate. https://www.desmos.com/calculator/pme3olhvn7

This is the limiting sum if a sequence is extended infinitely to the negative integers.

it has no meaning in the context of geometric sequences since the domain of a sequence is only positive integers.

Mixed Practice

FOUNDATION

1 Find x if each triple of three numbers form:

a an AP

i	$4, x, 16$	**ii**	$1, x, 49$	**iii**	$16, x, 25$
	10		25		20.5
iv	$-5, x, -20$	**v**	$x, 10, 50$	**vi**	$x, 12, 24$
	-12.5		-30		0
vii	$x, -1, 1$	**viii**	$x, 6, -12$	**ix**	$20, 30, x$
	-3		24		40
x	$-36, 24, x$	**xi**	$-\frac{1}{4}, -3, x$	**xii**	$7, -7, x$
	84		$-5\frac{3}{4}$		-21

b a GP

i	$4, x, 16$	**ii**	$1, x, 49$	**iii**	$16, x, 25$
	8 or -8		7 or -7		20 or -20
iv	$-5, x, -20$	**v**	$x, 10, 50$	**vi**	$x, 12, 24$
	10 or -10		2		6
vii	$x, -1, 1$	**viii**	$x, 6, -12$	**ix**	$20, 30, x$
	1		-3		45
x	$-36, 24, x$	**xi**	$-\frac{1}{4}, -3, x$	**xii**	$7, -7, x$
	-16		-36		7

2 Which term of the sequence 15, 24, 33, ... is 276?

$a = 15, d = 9$

$a_n = a + (n-1)d$

$276 = 15 + (n-1)9$

$261 = 9(n-1)$

$29 = n - 1$

$\therefore n = 30$

3 Which term of the sequence 25, 18, 11, ... is equal to −73?

$a = 25, d = -7$

$a_n = a + (n-1)d$

$-73 = 25 + (n-1)(-7)$

$-98 = -7(n-1)$

$14 = n - 1$

$\therefore n = 15$

4 Is 270 a term of the sequence 3, 11, 19, ...?

$a = 3, d = 8$

$a_n = a + (n-1)d$

Let $270 = 3 + (n-1)8$

$267 = 8(n-1)$

$n - 1 = 33.375$

Since n is not an integer, 270 is not a term.

5 Which term of the sequence 4, 20, 100, ... is equal to 12 500?

$a = 4, r = 5$

$a_n = ar^{n-1}$

$12500 = 4(5)^{n-1}$

$3125 = 5^{n-1}$

$5^5 = 5^{n-1}$

$\therefore n = 6$

6 Is 1200 a term of the sequence 2, 16, 128, ...?

$a = 2, r = 8$

Test if 1200 is a term:

$a_n = ar^{n-1}$

$1200 = 2(8)^{n-1}$

$600 = 8^{n-1}$

$\log_8 600 = n - 1$

$n - 1 = 3.076$

$n = 4.076$

$\therefore$ No

7 Which term of 3, 21, 147, ... is equal to 352 947?

$a = 3, r = 7$

$a_n = ar^{n-1}$

$352947 = 3(7)^{n-1}$

$117649 = 7^{n-1}$

$7^6 = 7^{n-1}$

$\therefore n = 7$

8 The first term of a geometric sequence is 7 and the 6th term is 1701. Find the common ratio.

$a = 7, a_6 = 1701$

$a_6 = ar^5$

$1701 = 7r^5$

$243 = r^5$

$r^5 = 3^5$

$\therefore r = 3$

9 A geometric series has limiting sum 6 and common ratio $\frac{1}{3}$.
Evaluate the first term of the series.

$S_\infty = 6, r = \frac{1}{3}$

$S_\infty = \frac{a}{1 - r}$

$6 = \frac{a}{1 - \frac{1}{3}}$

$6 = \frac{a}{\frac{2}{3}}$

$6 = \frac{3a}{2}$

$\therefore a = 4$

10 A geometric series has a limiting sum of 5 and first term 3. Find the common ratio.

$$S_\infty = 5, a = 3$$
$$S_\infty = \frac{a}{1-r}$$
$$5 = \frac{3}{1-r}$$
$$5 - 5r = 3$$
$$2 = 5r$$
$$\therefore r = \frac{2}{5}$$

11 Find the first six terms of the AP with first term $a = 7$ and sixth term $a_6 = 42$.

$$a_6 = a + 5d$$
$$42 = 7 + 5d$$
$$35 = 5d$$
$$d = 7$$
$$\therefore 7, 14, 21, 28, 35, 42$$

12 Find the first four terms of the GP with first term $a = 27$ and fourth term $a_4 = 8$.

$$a_4 = ar^3$$
$$8 = 27r^3$$
$$r^3 = \frac{8}{27}$$
$$r = \frac{2}{3}$$
$$\therefore 27, 18, 12, 8$$

13 Find the first five terms of the AP with $a = 48$ and $a_5 = 3$.

$$a_5 = a + 4d$$
$$3 = 48 + 4d$$
$$-45 = 4d$$
$$d = -11.25$$
$$\therefore 48, 36.75, 25.5, 14.25, 3$$

14 Find the first five terms of the GP with $a = 48$ and $a_5 = 3$.

$$a_5 = ar^4$$
$$3 = 48r^4$$
$$r^4 = \frac{1}{16}$$
$$r = \frac{1}{2}\left(or\ r = -\frac{1}{2}\right)$$
$$\text{For } r = \frac{1}{2}: 48, 24, 12, 6, 3$$
$$\text{For } r = -\frac{1}{2}: 48, -24, 12, -6, 3$$

15 Find the first term and common difference of the AP with $a_{10} = 18$ and $a_{20} = 48$.

$$a_10 = a + 9d = 18 \ldots (1)$$
$$a_20 = a + 19d = 48 \ldots (2)$$
$$(2) - (1): 10d = 30$$
$$d = 3$$
$$\text{Substitute into (1): } a + 27 = 18$$
$$\therefore a = -9, d = 3$$

16 Find the first term and common difference of the AP with $a_5 = 24$ and $a_9 = -12$.

$a + 4d = 24 \ldots (1)$

$a + 8d = -12 \ldots (2)$

$(2) - (1): 4d = -36$

$d = -9$

Substitute into (1): $a - 36 = 24$

$\therefore a = 60, d = -9$

17 Find the first term and common difference of the AP with $a_4 = 6$ and $a_{12} = 34$.

$a + 3d = 6 \ldots (1)$

$a + 11d = 34 \ldots (2)$

$(2) - (1): 8d = 28$

$d = 3.5$

Substitute into (1): a + 10.5 = 6

$\therefore a = -4.5, d = 3.5$

18 Find the first term and common ratio of the GP with $a_3 = 16$ and $a_6 = 128$.

$a_3 = ar^2 = 16 \ldots (1)$

$a_6 = ar^5 = 128 \ldots (2)$

$(2) \div (1): r^3 = 8$

$r = 2$

Substitute into (1): $4a = 16$

$\therefore a = 4, r = 2$

19 Find the first term and common ratio of the GP with $a_2 = \frac{1}{3}$ and $a_6 = 27$.

$ar = \frac{1}{3} \ldots (1)$

$ar^5 = 27 \ldots (2)$

$(2) \div (1): r^4 = 81$

$r^2 = 9$

$r = 3 \text{ or } -3$

For $r = 3: a = \frac{1}{9}$

For $r = -3: a = -\frac{1}{9}$

20 Find the first term and common ratio of the GP with $a_5 = 6$ and $a_9 = 24$.

$ar^4 = 6 \ldots (1)$

$ar^8 = 24 \ldots (2)$

$(2) \div (1): r^4 = 4$

$r^2 = 2$

$r = \sqrt{2} \text{ or } -\sqrt{2}$

For $r = \sqrt{2}: a(\sqrt{2})^4 = 6, 4a = 6, a = \frac{3}{2}$

For $r = -\sqrt{2}: a = \frac{3}{2}$

21 The third term of an AP is 7 and the seventh term is 31. Find the eighth term.

$$a + 2d = 7 \ldots (1)$$
$$a + 6d = 31 \ldots (2)$$
$$(2) - (1): 4d = 24$$
$$d = 6$$
$$a_8 = a_7 + d = 31 + 6 = 37$$

22 The common difference of an AP is −7 and the 10th term is 3. Find the second term.

$$a_{10} = a + 9d$$
$$3 = a + 9(-7)$$
$$3 = a - 63$$
$$a = 66$$
$$a_2 = a + d = 66 - 7 = 59$$

23 The common ratio of a GP is 2 and the sixth term is 6. Find the second term.

$$a_6 = ar^5$$
$$6 = a(2)^5$$
$$6 = 32a$$
$$a = \frac{3}{16}$$
$$a_2 = ar = \left(\frac{3}{16}\right)(2) = \frac{3}{8}$$

24 Find the first value of n for which the terms of the sequence 100, 93, 86, ... is less than 20.

$$a = 100, d = -7$$
$$a_n < 20$$
$$100 + (n - 1)(-7) < 20$$
$$100 - 7n + 7 < 20$$
$$107 - 7n < 20$$
$$87 < 7n$$
$$n > 12.43$$
$$\therefore n = 13$$

25 Find the values of n for which the terms of the sequence −86, −83, −80, ... are positive.

$$a = -86, d = 3$$
$$a_n > 0$$
$$-86 + (n - 1)3 > 0$$
$$-86 + 3n - 3 > 0$$
$$3n > 89$$
$$n > 29.67$$

When $n = 31, 32, 33 \ldots$ the sequence is positive

26 Find the first negative term of the sequence 54, 50, 46, ...

$$a = 54, d = 50 - 54 = -4$$
$$a_n < 0$$
$$54 + (n - 1)(-4) < 0$$
$$54 - 4n + 4 < 0$$
$$58 < 4n$$
$$n > 14.5$$
$$\therefore n = 15, \text{so } a_{15} = 54 + 14(-4)$$
$$= -2$$

27 Find the first term that is greater than 100 in the sequence 3, 7, 11, ...

$a = 3, d = 7 - 3 = 4$

$a_n > 100$

$3 + (n - 1)4 > 100$

$3 + 4n - 4 > 100$

$4n > 101$

$n > 25.25$

$\therefore n = 26, \text{so } a_{26} = 3 + 25(4) = 103$

28

a Show that $\sqrt{3}, \sqrt{12}, \sqrt{27}$, ... is an arithmetic sequence.

$\sqrt{3}, \sqrt{12}, \sqrt{27}$

$= \sqrt{3}, 2\sqrt{3}, 3\sqrt{3}$

Common difference $a_3 - a_2 = a_2 - a_1 = d = \sqrt{3}$, so sequence is AP

b Find the 50th term in simplest form.

$a = \sqrt{3}, d = \sqrt{3}$

$a_{50} = a + 49d$

$= \sqrt{3} + 49\sqrt{3}$

$= 50\sqrt{3}$

29 Which term of the sequence 8, −4, 2, ... is $\frac{1}{128}$?

$a = 8, r = -\frac{1}{2}$

$a_n = ar^{n-1}$

$\frac{1}{128} = 8\left(-\frac{1}{2}\right)^{n-1}$

$\frac{1}{1024} = \left(-\frac{1}{2}\right)^{n-1}$

$\left(\frac{1}{2}\right)^{10} = \left(-\frac{1}{2}\right)^{n-1}$

For this to be true, n − 1 must be even and $\left(\frac{1}{2}\right)^{n-1} = \left(\frac{1}{2}\right)^{10}$

$\therefore n - 1 = 10$

$\therefore n = 11$

30 Which term of 54, 18, 6, ... is $\frac{2}{243}$?

$a = 54, r = \frac{1}{3}$

$a_n = ar^{n-1}$

$\frac{2}{243} = 54\left(\frac{1}{3}\right)^{n-1}$

$\frac{1}{6561} = \left(\frac{1}{3}\right)^{n-1}$

$\frac{1}{3^8} = \left(\frac{1}{3}\right)^{n-1}$

$\therefore n - 1 = 8$

$\therefore n = 9$

31 The common ratio of a geometric series is 4 and the sum of the first 5 terms is 3069.
Find the first term.

$$r = 4, S_5 = 3069$$
$$S_5 = \frac{a(r^5 - 1)}{r - 1}$$
$$3069 = \frac{a(4^5 - 1)}{4 - 1}$$
$$3069 = \frac{a(1024 - 1)}{3}$$
$$3069 = \frac{a(1023)}{3}$$
$$3069 = 341a$$
$$\therefore a = 9$$

32 The limiting sum of a geometric series is $9\frac{1}{3}$ and the common ratio is $\frac{2}{5}$.
Find the first term of the series.

$$S_\infty = \frac{28}{3}, r = \frac{2}{5}$$
$$S_\infty = \frac{a}{1 - r}$$
$$\frac{28}{3} = \frac{a}{1 - \frac{2}{5}}$$
$$\frac{28}{3} = \frac{a}{\frac{3}{5}}$$
$$\frac{28}{3} = \frac{5a}{3}$$
$$\therefore a = \frac{28}{5}$$

33 A geometric series has limiting sum 40 and its first term is 5.
Find the common ratio of the series.

$$S_\infty = 40, a = 5$$
$$S_\infty = \frac{a}{1 - r}$$
$$40 = \frac{5}{1 - r}$$
$$40(1 - r) = 5$$
$$40 - 40r = 5$$
$$35 = 40r$$
$$\therefore r = \frac{7}{8}$$

34 A frog jumps 0.5 metres. On each subsequent jump it travels 0.2 of the previous distance.
Find the total distance through which the frog jumps.

$$a = 0.5$$
$$r = 0.2$$
$$S_\infty = \frac{a}{1 - r}$$
$$= \frac{0.5}{1 - 0.2}$$
$$= \frac{0.5}{0.8}$$
$$= 0.625 \text{ m}$$

35 A tree enters a phase where it grows by $\frac{4}{5}$ of each previous year's growth.
It is initially 3 m high, what is its ultimate height?

$$a = 3, r = \frac{4}{5}$$

$$s_\infty = \frac{3}{1-\frac{4}{5}} = 15 \text{ m}$$

36 Find the nth term of the sequences:

a

$$\frac{2}{1}, \frac{2+4}{1+3}, \frac{2+4+6}{1+3+5}, \cdots$$

Numerators: AP where $a = 2, d = 2$:

$$n\text{th sum} = \frac{n}{2(2\times 2 + (n-1)2)} = n(n+1)$$

Denominators: AP where $a = 1, d = 2$:

$$n\text{th sum} = \frac{n}{2(2\times 1 + (n-1)2)} = n^2$$

$$a_n = \frac{n(n+1)}{n^2} = \frac{n+1}{n}$$

b

$$\frac{1}{1}, \frac{1+2}{1+4}, \frac{1+2+4}{1+4+16}, \cdots$$

Numerators: GP where $a = 1, r = 2$:

$$n\text{th sum} = \frac{2^n-1}{2-1} = 2^n - 1 = 2^n - 1$$

Denominators: GP where $a = 1, r = 4$:

$$n\text{th sum} = \frac{4^n-1}{4-1} = \frac{4^n-1}{3}$$

$$a_n = \frac{2^n - 1}{\left(\frac{4^n-1}{3}\right)} = \frac{3(2^n-1)}{4^n-1} = \frac{3(2^n-1)}{2^{2n}-1} = \frac{3}{2^n+1}$$

37 Find a and d for the AP in which $a_6 + a_8 = 44$ and $a_{10} + a_{13} = 35$.

$$(a + 5d) + (a + 7d) = 44$$
$$2a + 12d = 44$$
$$a + 6d = 22 \ldots (1)$$
$$(a + 9d) + (a + 12d) = 35$$
$$2a + 21d = 35 \ldots (2)$$
$$(2) - 2(1): (2a + 21d) - 2(a + 6d) = 35 - 44$$
$$9d = -9$$
$$d = -1$$
$$\text{Substitute into (1): } a - 6 = 22$$
$$\therefore a = 28, d = -1$$

38 Find a and r for the GP in which $a_2 + a_3 = 4$ and $a_4 + a_5 = 36$.

$$ar + ar^2 = 4$$
$$ar(1 + r) = 4 \ldots (1)$$
$$ar^3 + ar^4 = 36$$
$$ar^3(1 + r) = 36 \ldots (2)$$
$$(2) \div (1): r^2 = 9$$
$$r = 3 \text{ or } -3$$
$$\text{For } r = 3: a(3)(4) = 4, a = \frac{1}{3}$$
$$\text{For } r = -3: a(-3)(-2) = 4, a = \frac{2}{3}$$
$$\therefore a = \frac{1}{3}, r = 3 \text{ or } a = \frac{2}{3}, r = -3$$

39 The fourth, sixth and eighth terms of an AP add to −6. Find the sixth term.

$$a_4 + a_6 + a_8 = -6$$
$$(a + 3d) + (a + 5d) + (a + 7d) = -6$$
$$3a + 15d = -6$$
$$a + 5d = -2$$
$$\therefore a_6 = a + 5d = -2$$

40 The sum of the first 5 terms of an arithmetic sequence is 110 and the sum of the first 10 terms is 320. Find the first term and the common difference.

$$S_5 = 110, S_{10} = 320$$
$$S_5 = \frac{5}{2}[2a + 4d] = 110$$
$$5(2a + 4d) = 220$$
$$2a + 4d = 44$$
$$a + 2d = 22 \ldots (1)$$
$$S_{10} = \frac{10}{2}[2a + 9d] = 320$$
$$5(2a + 9d) = 320$$
$$2a + 9d = 64 \ldots (2)$$
$$(2) - 2(1): (2a + 9d) - 2(a + 2d) = 64 - 44$$
$$5d = 20$$
$$d = 4$$
$$\text{Substitute into (1): } a + 8 = 22$$
$$\therefore a = 14, d = 4$$

41 The sum of the first 5 terms of an arithmetic sequence is 35 and the sum of the next 5 terms is 160. Find the first term and the common difference.

$$S_5 = 35, \text{sum of next 5 terms} = 160$$
$$S_5 = \frac{5}{2}[2a + 4d] = 35$$
$$2a + 4d = 14$$
$$a + 2d = 7 \ldots (1)$$
$$\text{Sum of terms 6 to 10} = S_{10} - S_5 = 160$$
$$S_{10} = 195$$
$$S_{10} = \frac{10}{2}[2a + 9d] = 195$$
$$2a + 9d = 39 \ldots (2)$$
$$(2) - 2(1): 5d = 25$$
$$d = 5$$
$$\text{Substitute into (1): } a + 10 = 7$$
$$\therefore a = -3, d = 5$$

42 The 3rd term of a geometric sequence is $\frac{2}{5}$ and the 5th term is $\frac{8}{5}$.
Find the first term and common ratio.

$$ar^2 = \frac{2}{5} \ldots (1), \quad ar^4 = \frac{8}{5} \ldots (2)$$
$$(2) \div (1): r^2 = 4$$
$$r = 2 \text{ or } -2$$
$$\text{For } r = 2: a(4) = \frac{2}{5}, a = \frac{1}{10}$$
$$\text{For } r = -2: a(4) = \frac{2}{5}, a = \frac{1}{10}$$

43 How many terms of the series $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+$ ... give a sum of $\frac{1023}{1024}$?

$$a=\frac{1}{2}, r=\frac{1}{2}, S_n=\frac{1023}{1024}$$

$$S_n=\frac{a(1-r^n)}{1-r}$$

$$\frac{1023}{1024}=\frac{\left(\frac{1}{2}\right)\left(1-\left(\frac{1}{2}\right)^n\right)}{1-\frac{1}{2}}$$

$$\frac{1023}{1024}=1-\left(\frac{1}{2}\right)^n$$

$$\left(\frac{1}{2}\right)^n=\frac{1}{1024}$$

$$\left(\frac{1}{2}\right)^n=\left(\frac{1}{2}\right)^{10}$$

$$\therefore n=10 \text{ terms}$$

44 The second term of a geometric series is 2 and its limiting sum is 9.
Find the values of first term a and common ratio r.

$$a_2=2, S_\infty=9$$

$$ar=2 \ldots (1)$$

$$\frac{a}{1-r}=9 \ldots (2)$$

$$\text{From (1): } a=\frac{2}{r}$$

$$\text{Substitute into (2): } \frac{\frac{2}{r}}{1-r}=9$$

$$\frac{2}{r(1-r)}=9$$

$$2=9r-9r^2$$

$$9r^2-9r+2=0$$

$$(3r-2)(3r-1)=0$$

$$r=\frac{2}{3} \text{ or } \frac{1}{3}$$

$$\text{For } r=\frac{2}{3}: a=\frac{2}{\frac{2}{3}}=3$$

$$\text{For } r=\frac{1}{3}: a=\frac{2}{\frac{1}{3}}=6$$

45 How many terms of the series 80 + 73 + 66 + ... are needed to give a sum of 495?

$$a=80, d=73-80=-7, S_n=495$$

$$S_n=\frac{n}{2}[2a+(n-1)d]$$

$$495=\frac{n}{2}[2(80)+(n-1)(-7)]$$

$$990=n[160-7n+7]$$

$$990=n(167-7n)$$

$$990=167n-7n^2$$

$$7n^2-167n+990=0$$

$$(7n-90)(n-11)=0$$

$$n=\frac{90}{7} \text{ or } 11$$

Since n must be positive integer: $n=11$

46 We can convert recurring decimals to fractions by using the limiting sum of a GP.

Worked Example	Your Turn
$0.\overline{27}$ $0.\overline{27} = 0.2727272727272727\ldots$ $= 0.27 + 0.0027 + 0.000027 + \cdots$ This is a GP with $a = 0.27, r = 0.01$ $0.\overline{27} = S_\infty = \frac{0.27}{1 - 0.01} = \frac{3}{11}$	$0.\dot{3}$ $0.\bar{3} = 0.333333\ldots$ $= 0.3 + 0.03 + 0.003 + \ldots$ This is a GP with $a = 0.3, r = 0.1$ $0.\bar{3} = S_\infty = \frac{0.3}{1 - 0.1} = \frac{1}{3}$
$2.64\dot{5}$ $2.64\dot{5} = 2.64555555555555555555\ldots$ $= 2.64 + 0.005 + 0.0005 + 0.00005\ldots$ Starting from 0.005, this is a GP with $a = 0.27, r = 0.1$ $2.64\dot{5} = 2.64 + S_\infty$ $= 2.64 + \frac{0.005}{1 - 0.1}$ $= \frac{2381}{900}$	$7.\overline{81}$ $7.\overline{81} = 7.818181\ldots$ $= 7 + 0.81 + 0.0081 + 0.000081 + \ldots$ Starting from 0.81, this is a GP with $a = 0.81, r = 0.01$ $7.\overline{81} = 7 + S_\infty$ $= 7 + \frac{0.81}{1 - 0.01}$ $= \frac{86}{11}$

47 When light passes through one sheet of very thin glass, its intensity is reduced by 3%.

a If the light passes through 50 sheets of this glass, find by what percentage the intensity will be reduced.

Intensity after each sheet = 0.97 × previous

After 50 sheets: $I = I_0(0.97)^{50} = 0.218\, I_0$

Percentage reduction = 100% − 21.8% = 78.2%

∴ about 78%

b What is the minimum number of sheets that will reduce the intensity below 1%?

Find n where $(0.97)^n < 0.01$

$n > \log_{0.97} 0.01$

$n > 151.5$

∴ 152 sheets

Note: flip the inequality when taking logarithm with base between 0 and 1.

48 How many terms of the series 20 + 18 + 16 + ... are needed to give a sum of 104?

$a = 20, d = 18 - 20 = -2, S_n = 104$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$104 = \frac{n}{2}[2(20) + (n-1)(-2)]$

$208 = n[40 - 2n + 2]$

$208 = n(42 - 2n)$

$208 = 42n - 2n^2$

$2n^2 - 42n + 208 = 0$

$n^2 - 21n + 104 = 0$

$(n-13)(n-8) = 0$

$n = 13 \text{ or } 8$

$\therefore$ 8 terms or 13 terms

49 The sum of 12 terms of an arithmetic series is 186 and the 20th term is 83. Find the sum of 40 terms of the series.

$S_{12} = 186, a_{20} = 83$

$S_{12} = \frac{12}{2}[2a + 11d] = 186$

$6(2a + 11d) = 186$

$2a + 11d = 31 \ldots (1)$

$a_{20} = a + 19d = 83 \ldots (2)$

From (2): $a = 83 - 19d$

Substitute into (1): $2(83 - 19d) + 11d = 31$

$166 - 38d + 11d = 31$

$-27d = -135$

$d = 5$

From (2): $a = 83 - 95 = -12$

$S_{40} = \frac{40}{2}[2(-12) + 39(5)]$

$= 3420$

50 The sum of the first 4 terms of an arithmetic series is 42 and the sum of the 3rd and 7th term is 46. Find the sum of the first 20 terms.

$S_4 = 42, a_3 + a_7 = 46$

$S_4 = \frac{4}{2}[2a + 3d] = 42$

$2(2a + 3d) = 42$

$2a + 3d = 21 \ldots (1)$

$a_3 + a_7 = (a + 2d) + (a + 6d) = 46$

$2a + 8d = 46$

$a + 4d = 23 \ldots (2)$

From (2): $a = 23 - 4d$

Substitute into (1): $2(23 - 4d) + 3d = 21$

$46 - 8d + 3d = 21$

$-5d = -25$

$d = 5$

From (2): $a = 23 - 20 = 3$

$S_{20} = \frac{20}{2}[2(3) + 19(5)]$

$= 1010$

51

a Show that $x + 1, 2x + 4, 3x + 7$, ... are the first 3 terms in an arithmetic sequence.

$(2x + 4) - (x + 1) = x + 3$

$(3x + 7) - (2x + 4) = x + 3$

$\therefore$ Common difference $d = x + 3$, so sequence is AP

b Find an expression for the sum of the first 50 terms of the sequence.

$a = x + 1, d = x + 3$

$S_{50} = \frac{50}{2}[2a + 49d]$

$= 25[2(x + 1) + 49(x + 3)]$

$= 25[2x + 2 + 49x + 147]$

$= 25[51x + 149]$

$= 25(51x + 149)$

52 The 20th term of an arithmetic series is 131 and the sum of the 6th to 10th terms inclusive is 235. Find the sum of the first 20 terms.

$a_{20} = 131$, sum of $a_6 to\ a_{10} = 235$

$a_{20} = a + 19d = 131 \ldots (1)$

Sum of a_6 to $a_{10} = a_6 + a_7 + a_8 + a_9 + a_{10}$

$= (a + 5d) + (a + 6d) + (a + 7d) + (a + 8d) + (a + 9d)$

$= 5a + 35d = 235$

$a + 7d = 47 \ldots (2)$

$(1) - (2): 12d = 84$

$d = 7$

From (2): $a + 49 = 47$

$a = -2$

$S_{20} = \frac{20}{2}[2(-2) + 19(7)]$

$= 10(-4 + 133)$

$= 10(129)$

$= 1290$

53 Find the value of n for the first term of the sequence 5000, 1000, 200, ... that is less than 1.

$a = 5000, r = \frac{1}{5}$

$a_n < 1$

$5000\left(\frac{1}{5}\right)^{n-1} < 1$

$\left(\frac{1}{5}\right)^{n-1} < \frac{1}{5000}$

$n - 1 > \log_{\frac{1}{5}}\left(\frac{1}{5000}\right)$

$n - 1 > 5.29$

$n > 6.29$

$\therefore n = 7$

54 Find the number of terms needed to be added for the sum to exceed 1 000 000 in the series 4 + 16 + 64 + ...

$a = 4, r = 4$

$S_n > 1000000$

$\frac{a(r^n - 1)}{r - 1} > 1000000$

$\frac{4(4^n - 1)}{4 - 1} > 1000000$

$4^n - 1 > 750000$

$4^n > 750001$

$n > \log_4 750001$

$n > 9.76$

$\therefore$ 10 terms

55 A geometric series has 3rd term 12 and 4th term −3. Find a, r and the limiting sum.

$a_3 = 12, a_4 = -3$

$ar^2 = 12 \ldots (1)$

$ar^3 = -3 \ldots (2)$

$(2) \div (1): r = -\frac{1}{4}$

Substitute into $(1): a\left(\frac{1}{16}\right) = 12$

$\therefore a = 192$

$S_\infty = \frac{a}{1 - r}$

$= \frac{192}{1 - \left(-\frac{1}{4}\right)}$

$= \frac{768}{5}$

$\therefore a = 192, r = -\frac{1}{4}, S_\infty = \frac{768}{5}$

56 The price of shares in a particular company is falling by an average of 2% each day.

a What percentage of their initial value do they have after 2 days?

GP where each day value = 0.98 × previous day

After 2 days: $(0.98)^2 = 0.9604 = 96.04\%$

b Approximately how many days will it take for the shares to halve in value?

$(0.98)^n = 0.5$

$n = \log_{0.98} 0.5$

n = 34.3

$\therefore$ 34 days

57 A weight on an elastic string drops down 60 cm and then bounces back to $\frac{2}{3}$ of its initial height. It keeps bouncing, each time rising back to $\frac{2}{3}$ of its previous height. What is the total distance through which the weight travels?

Weight drops 60 cm, bounces back to 40 cm

First up and down distance is 100 cm

Each bounce is $\frac{2}{3}$ of previous height, so distances form GP with $a = 100, r = \frac{2}{3}$

$S_\infty = \frac{100}{1 - \frac{2}{3}} = 300$ cm

58 Mary bounces a ball, dropping it from 1.5 m on its first bounce.

It then rises up to $\frac{2}{5}$ of its height on each bounce.

Find the distance through which the ball travels.

Ball drops 1.5 m, bounces back to 0.6 m

First up and down distance is 2.1 m

Each bounce is $\frac{2}{5}$ of previous height, so distances form GP with $a = 2.1, r = \frac{2}{5}$

$$S_\infty = \frac{2.1}{1-\frac{2}{5}} = 3.5 \text{ m}$$

59 A market gardener plants daffodil bulbs in rows, starting with a row of 45 bulbs. Each successive row has 5 more bulbs than the row before.

a Calculate the number of bulbs in the 34th row.

$$a = 45, d = 5$$
$$a_{34} = a + 33d$$
$$= 45 + 33(5)$$
$$= 210$$

b Which row would be the first to have more than 100 bulbs in it?

$$a_n > 100$$
$$45 + (n-1)5 > 100$$
$$45 + 5n - 5 > 100$$
$$5n > 60$$
$$n > 12$$

∴ 13th row

c The market gardener plants 10 545 bulbs altogether. How many rows are there?

$$a = 45, d = 5, S_n = 10545$$
$$S_n = \frac{n}{2}[2a + (n-1)d]$$
$$10545 = \frac{n}{2}[2(45) + (n-1)5]$$
$$21090 = n[90 + 5n - 5]$$
$$21090 = n(85 + 5n)$$
$$21090 = 85n + 5n^2$$
$$5n^2 + 85n - 21090 = 0$$
$$n^2 + 17n - 4218 = 0$$

Using quadratic formula:

$$n = 57 \text{ or } -74$$

∴ 57 rows

18 Show that $\log_2 96, \log_2 24, \log_2 6$, ... is an AP, and show that $a_n = 7 - 2n + \log_2 3$.

$$a_2 - a_1 = \log_2 24 - \log_2 96 = \log_2\left(\frac{24}{96}\right) = \log_2\left(\frac{1}{4}\right) = -2$$
$$a_3 - a_2 = \log_2 6 - \log_2 24 = \log_2\left(\frac{6}{24}\right) = \log_2\left(\frac{1}{4}\right) = -2$$

Common difference $d = -2$ ∴ AP

$$a_n = a + (n-1)d$$
$$= (5 + \log_2 3) + (n-1)(-2)$$
$$= 5 + \log_2 3 - 2n + 2$$
$$= 7 - 2n + \log_2 3$$

60 A stack of logs has 1 on the top, then 3 on the next row down, and each successive row has 2 more logs than the one on top of it.

a How many logs are in the 20th row?

$$a = 1, d = 2$$
$$a_{20} = a + 19d$$
$$= 1 + 19(2)$$
$$= 39$$

b Which row has 57 logs?

$$a_n = 57$$
$$1 + (n - 1)2 = 57$$
$$1 + 2n - 2 = 57$$
$$2n = 58$$
$$\therefore n = 29$$

c If there are 1024 logs altogether, how many rows are in the stack?

$$a = 1, d = 2, S_n = 1024$$
$$S_n = \frac{n}{2}[2a + (n - 1)d]$$
$$1024 = \frac{n}{2}[2(1) + (n - 1)2]$$
$$2048 = n[2 + 2n - 2]$$
$$2048 = 2n^2$$
$$n^2 = 1024$$
$$\therefore n = 32 \text{ rows}$$

61 A sculpture consists of a set of poles set in a row, with the tallest pole 2.4 m high, the next pole 2.1, the next one 1.8 and so on, down to the last pole which is 0.6 m high.

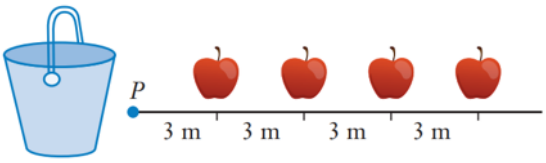
a How many poles are in the sculpture?

$$a = 2.4,\ a_n = 0.6,\ d = -0.3$$
$$a_n = a + (n - 1)d$$
$$0.6 = 2.4 + (n - 1)(-0.3)$$
$$0.6 = 2.4 - 0.3n + 0.3$$
$$0.6 = 2.7 - 0.3n$$
$$0.3n = 2.1$$
$$\therefore n = 7 \text{ poles}$$

b The poles are made of timber. What length of timber is there altogether in the poles?

$$S_7 = \frac{7}{2}[2.4 + 0.6]$$
$$= \frac{7}{2}(3)$$
$$= 10.5 \text{ m}$$

62 A child starts at a bucket placed on the ground and runs and picks up an apple 3 m away.
She then runs back to the bucket and puts the apple in a bucket.
The child then runs to get the next apple 6 m away, and runs back to to place it in the bucket.
This continues until she has all the apples in the bucket.
The apples are placed 3 metres apart from each other.



a How far does the child run from the bucket to pick up the kth apple?

Apples are at: 3, 6, 9, 12, ... m

AP where a = 3, d = 3

$a_k = 3k$ m

b How far does the child run to fetch all k apples, including return trips to the bucket?

Each apple requires two trips (there and back):

For kth apple: $2(3k) = 6k$ m

Total for k apples:

$= 6(1) + 6(2) + 6(3) + \ldots + 6(k)$

$= 6(1 + 2 + 3 + \ldots + k)$

$= 6 \times \frac{k(k+1)}{2}$

$= 3k(k+1)$ m

c The child runs 270 m to fetch all the apples and return them to the bucket.
How many apples are there?

$3k(k+1) = 270$

$k(k+1) = 90$

$k^2 + k - 90 = 0$

$(k+10)(k-9) = 0$

$k = 9 \text{ or } -10$

$\therefore$ 9 apples

Exam Questions

1 2019 HSC Advanced Band 3

What is the limiting sum of the following geometric series?

$$2000 - 1200 + 720 - 432...$$

$$S_\infty = \frac{a}{1-r}$$
$$= \frac{2000}{1+\frac{3}{5}}$$
$$= 1250$$

2 2024 HSC Advanced Band 3

Find the sum of the terms in the arithmetic series

50 + 57 + 64 + ... + 2024

$$a = 50, d = 7$$
$$\text{Last term} = a + (n-1)d$$
$$2024 = 50 + (n-1)d$$
$$n - 1 = \frac{2024 - 50}{7}$$
$$n = 283$$
$$S_{283} = \frac{n}{2}(a + a_{283})$$
$$= \left(\frac{283}{2}\right)(50 + 2024)$$
$$= 293\,471$$

3 HSC Sample Question Advanced Band 3

On day 1, Vikki spends 90 minutes on a training program.

On each following day, she spends 10 minutes less on the training program than she did the day before.

Let t_n be the number of minutes that Vikki spends on the training program on day n.

A recursive equation that can be used to model this situation for $1 \leq n \leq 10$ is

A. $t_{n+1} = 0.90t_n \quad t_1 = 90$

B. $t_{n+1} = 1.10t_n \quad t_1 = 90$

C. $t_{n+1} = 1 - 10t_n \quad t_1 = 90$

D. $t_{n+1} = t_n - 10 \quad t_1 = 90$

Each term is 10 minutes less than the previous term.

D

4 2005 HSC Advanced Band 3

Evaluate

$$\sum_{n=3}^{5}(2n+1)$$

$$(2(3) + 1) + (2(4) + 1) + (2(5) + 1)$$
$$= 7 + 9 + 11$$
$$= 27$$

5 HSC Sample Question Advanced Band 3

Joe buys a tractor under a buy-back scheme. This scheme gives Joe the right to sell the tractor back to the dealer.

The recurrence relation below can be used to calculate the price Joe sells the tractor back to the dealer (P_n), after n years

$P_0 = 56\,000, \quad P_n = P_{n-1} - 7000$

a Write the general rule to find the value of P_n in terms of n.

$a = 56000, d = -7000$

$P_n = 56\,000 - 7000n$

b After how many years will the dealer offer to buy back Joe's tractor at half of its original value.

$28\,000 = 56\,000 - 7000n$

$7000n = 28\,000$

$\therefore n = 4$ years

6 2006 HSC Advanced Band 3

On the first day of the harvest, an orchard produces 560 kg of fruit.
On the next day, the orchard produces 543 kg, and the amount produced continues to decrease by the same amount each day.

a How much fruit is produced on the fourteenth day of the harvest?
b What is the total amount of fruit that is produced in the first 14 days of the harvest?
c On what day does the daily production first fall below 60 kg?

AP where $a = 560, d = -17$

$\vdots$

$a_{14} = a + 13d$

$= 560 - (13)(17)$

$= 339$

$\therefore$ 339 kg of fruit produced on 14th day.

$S_{14} = \left(\frac{n}{2}\right)[2a + (n-1)d]$

$= \left(\frac{14}{2}\right)[2(560) - (14-1)(17)]$

$= 7[1120 - 221]$

$= 6293$

$\therefore$ 6293 kg of fruit produced in the 1st to 14th days.

$a_n = a + (n-1)d < 60$

$560 - 17(n-1) < 60$

$560 - 17n + 17 < 60$

$17n > 517$

$n > 30.41\ldots$

$\therefore$ On the 31st day, production will drop below 60 kg

7 2009 HSC Advanced Band 3

Evaluate

$$\sum_{k=1}^{4}(-1)^k k^2$$

$$= (-1)^1(1^2) + (-1)^2(2^2) + (-1)^3(3^2) + (-1)^4(4^2)$$

$$= -1 + 4 - 9 + 16$$

$$= 10$$

8 2025 HSC Advanced Band 3

The numbers, 75, p, q, 2025, form a geometric sequence.

Find the values of p and q.

$$a = 75,$$

$$75r = p,$$

$$75r^2 = q,$$

$$75r^3 = 2025$$

$$\text{Using } 75r^3 = 2025:$$

$$r = \sqrt[3]{\frac{2025}{75}} = 3$$

$$p = 75(3) = 225$$

$$q = 75(3^2) = 675$$

9 2023 HSC Advanced Band 4

The fourth term of a geometric sequence is 48.

The eighth term of the same sequence is $\frac{3}{16}$.

Find the possible value(s) of the common ratio and the corresponding first term(s)

$$T_4 = ar^3 = 48 \ldots (1)$$

$$T_8 = ar^7 = \left(\frac{3}{16}\right) \ldots (2)$$

$$\frac{(2)}{(1)} = \frac{ar^7}{ar^3} = \frac{\frac{3}{16}}{48}$$

$$r^4 = \frac{1}{256}$$

$$r = \pm\left(\frac{1}{4}\right)$$

$$\text{If } r = \left(\frac{1}{4}\right)$$

$$a\left(\frac{1}{4}\right)^3 = 48$$

$$\frac{a}{64} = 48$$

$$a = 3072$$

$$\text{If } r = -\left(\frac{1}{4}\right), a = -3072$$

10 HSC Sample Question Advanced Band 4

Phil is a builder who has purchased a large set of tools.

The value of Phil's tools is depreciated using the reducing balance method.

The value of the tools, in dollars, after n years, V_n, can be modelled by the recurrence relation shown below.

$$V_0 = 60\,000, \quad V_{n+1} = 0.9V_n$$

a Use recursion to find the value of the tools after two years.

$V_0 = 60\,000$

$V_1 = 0.9 \times 60\,000 = 54\,000$

$V_2 = 0.9 \times 54\,000 = \$48\,600$

b Phil plans to replace these tools when their value first falls below \$20 000.
After how many years will Phil replace these tools?

$60\,000(0.9)^n < 20\,000$

$0.9^n < \frac{1}{3}$

$n > \log_{0.9}\frac{1}{3}$

$n > 10.427$

$\therefore$ Phil will replace in the 11th year.

c Phil has another option for depreciation. He depreciates the value of the tools by a flat rate of 8% of the purchase price per annum.
Let V_n be the value of the tools after n years, in dollars.
Write down a recurrence relation, in terms of V_0, V_{n+1} and V_n, that could be used to model the value of the tools using this flat rate depreciation.

Flat rate means this is an AP.

Annual depreciation $= 0.08 \times V_0$

$\therefore V_{n+1} = V_n - 0.08V_0$

11 HSC Sample Question Advanced Band 4

Express the recurring decimal 0.323232... as a fraction.

$0.3232\ldots = 0.32 + 0.0032 + 0.000032\ldots$

$\Rightarrow GP$ where $a = 1, r = 0.01$

$S_\infty = \left(\frac{0.32}{1 - 0.01}\right)$

$= \frac{32}{99}$

12 2004 HSC Advanced Band 4

Evaluate

$$\sum_{n=2}^{4} n^2$$

$2^2 + 3^2 + 4^2$

$= 4 + 9 + 16$

$= 29$

13 2012 HSC Advanced Band 4

Rectangles of the same height are cut from a strip and arranged in a row. The first rectangle has width 10cm. The width of each subsequent rectangle is 96% of the width of the previous rectangle.

a Find the length of the strip required to make the first ten rectangles.

b Explain why a strip of 3m is sufficient to make any number of rectangles.



10 cm

$T_1 = a = 10$

$T_2 = ar = 10 \times 0.96 = 9.6$

$T_3 = ar^2 = 10 \times 0.96^2 = 9.216$

$\Rightarrow$ GP where $a = 10$ and $r = 0.96$

S_{10} = Length of strip for 10 rectangles

$= \frac{a(1-r^n)}{1-r}$

$= 10\left(\frac{1-0.96^{10}}{1-0.96}\right)$

= 83.8 cm (to 1 d.p.)

$S_\infty = \frac{a}{1-r}$

$= \frac{10}{1-0.96}$

= 250 cm

$\therefore$ Since 3 m > 2.5 m, it is sufficient.

14 HSC Sample Question Advanced Band 4

When placed in a pond, the length of a fish was 14.2 centimetres.

During its first month in the pond, the fish increased in length by 3.6 centimetres.

During its nth month, the fish increased in length by G_n centimetres, so $G_{n+1} = 0.75G_n$

Calculate the maximum length this fish can grow to.

Initial length = 14.2 cm

$G_1 = 3.6$

$G_2 = (0.75)G_1 = (0.75)3.6$

$G_3 = (0.75)G_2 = (0.75^2)3.6$

Growth is a geometric sequence

$3.6, 3.6(0.75), 3.6(0.75)^2, \ldots$

GP where $a = 3.6, r = 0.75$

Since $|r| < 1$,

$S_\infty = \frac{3.6}{1-0.75}$

$= 14.4$

$\therefore$ Maximum length of fish

= 14.2 + 14.4

= 28.6 cm

15 2008 HSC Advanced Band 4

The zoom function in a software package multiplies the dimensions of an image by 1.2. In an image, the height of a building is 50 mm. After the zoom function is applied once, the height of the building in the image is 60 mm. After the second application, it is 72 mm.

a Calculate the height of the building in the image after the zoom function has been applied eight times. Give your answer to the nearest mm.

b The height of the building in the image is required to be more than 400 mm. Starting from the original image, what is the least number of times the zoom function must be applied?

$T_1 = a = 50$

$T_2 = ar^1 = 50(1.2) = 60$

$T_3 = ar^2 = 50(1.2)^2 = 72$

GP where $a = 50, r = 1.2$

$T_9 = 50(1.2)^8$

= 214.99

∴ Height will be 215 mm (nearest mm)

$T_n = ar^{n-1} > 400$

$\therefore 50(1.2)^{n-1} > 400$

$1.2^{n-1} > 8$

$n-1 > \log_{1.2} 8$

$n > 12.405$

∴ The height of the building in the 13th image will be higher than 400 mm, which is the 12th time the zoom would be applied.

16 HSC Sample Question Advanced Band 4

There are 3000 tickets available for a concert.

On the first day of ticket sales, 200 tickets are sold.

On the second day, 250 tickets are sold.

On the third day, 300 tickets are sold.

This pattern of ticket sales continues until all 3000 tickets are sold.

How many days does it take for all of the tickets to be sold?

A. 5

B. 6

C. 8

D. 34

200 + 250 + 300 + ...

AP where $a = 200, d = 50$

Find n when $S_n = 3000$:

$S_n = \left(\frac{n}{2}\right)[2a + (n-1)d]$

$3000 = \frac{n}{2}[2(200) + (n-1)50]$

$= \frac{n}{2}(400 + 50n - 50)$

$= \frac{n}{2}(350 + 50n)$

$3000 = 175n + 25n^2$

$n^2 + 7n - 120 = 0$

$(n - 8)(n + 15) = 0$

$\therefore n = 8, (n > 0)$

C

17 2017 HSC Advanced Band 4

In an arithmetic series, the fifth term is 200 and the sum of the first four terms is 1200.

Find the value of the tenth term.

$T_n = a + (n-1)d$

$\Rightarrow a + 4d = 200 \ldots (1)$

$S_n = \left(\frac{n}{2}\right)[2a + (n-1)d]$

$\Rightarrow 4a + 6d = 1200 \ldots (2)$

$(1) \times 4 \Rightarrow 4a + 16d = 800 \ldots (3)$

$(3) - (2): 10d = -400$

$d = -40$

Substitute $d = -40$ into (1)

$a - 160 = 200$

$a = 360$

$\therefore T_{10} = 360 - 9(-40) = 0$

18 2014 HSC Advanced Band 4

Which expression is a term of the geometric series $3x - 6x^2 + 12x^3 - \ldots$?

A. $3072x^{10}$

B. $-3072x^{10}$

C. $3072x^{11}$

D. $-3072x^{11}$

$a = 3x, r = -2x$

$\therefore T_n = ar^n = 3x(-2x)^n$

$= 3(-2)^n x^{n+1}$

If $n = 9$

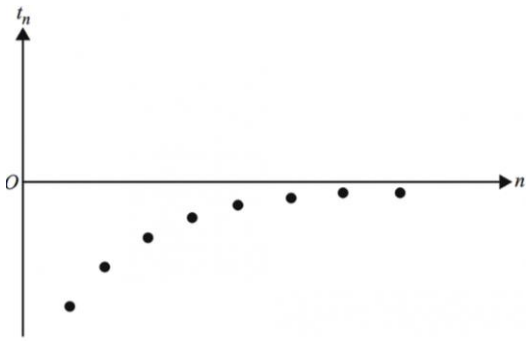
$T_9 = 3(-2)^9 x^{9+1} = -1536x^{10}$

If $n = 10$

$T_{10} = 3(-2)^{10} x^{10+1} = 3072x^{11}$

C

19 HSC Sample Question Advanced Band 5



The graph above shows consecutive terms of a sequence.

The sequence could be

A. geometric with common ratio r, where $r < 0$

B. geometric with common ratio r, where $0 < r < 1$

C. geometric with common ratio r, where $r > 1$

D. arithmetic with common difference d, where $d < 0$

Series is a GP with a limiting sum, so $0 < |r| < 1$

Terms do not oscillate between negative and positive, so C

20 HSC Sample Question Advanced Band 4

The first four terms of a geometric sequence are

4, −8, 16, −32

The sum of the first ten terms of this sequence is

A. -2048

B. -1364

C. 684

D. 1367

GP where $a = 4$

$r = -2$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$\therefore S_{10} = \frac{4[(-2)^{10} - 1]}{-2 - 1}$

$= -1364$

B

21 HSC Sample Question Advanced Band 4

A dragster is travelling at a speed of 100 km/h. It increases its speed by

50 km/h in the 1st second

30 km/h in the 2nd second

18 km/h in the 3rd second

and so on in this pattern.

Correct to the nearest whole number, the greatest speed, in km/h, that the dragster will reach is

A. 125

B. 200

C. 220

D. 225

Sequence of speed increases is

50, 30, 18,...

GP where $a = 50$, and $r = 0.6$

Since $|r| < 1$,

$S_\infty = \frac{a}{1 - r}$

$= \frac{50}{1 - 0.6}$

= 125

∴ Max speed is 100 + 125 = 225 km/h

⇒ D

22 HSC Sample Question Advanced Band 5

The first three terms of an arithmetic sequence are 1, 3, 5...

The sum of the first n terms of this sequence, S_n, is

A. $S_n = n^2$

B. $S_n = 2n$

C. $S_n = 2n - 1$

D. $S_n = 2n + 1$

AP where $a = 1$, and $d = 2$

$S_n = \frac{n}{2}[2a + (n - 1)d]$

$= \frac{n}{2}[2(1) + (n - 1)2]$

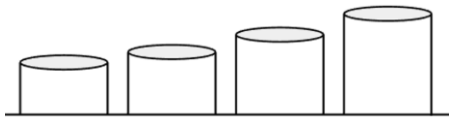
$= \frac{n}{2}[2 + 2n - 2]$

$= n^2$

A

23 HSC Sample Question Advanced Band 5

An army training fitness station requires soldiers to climb up steel cylinders steps that gradually ascend in height, as shown below.



The shortest cylinder is 0.7 metres high and the highest is 4.0 metres.
The height of each cylinder increases by the same amount and the fitness station is made up of a total of 23 steel cylinders.

a Determine the height of the 13th steel cylinder.

b Show that the total height of the steel cylinder steps in the fitness station is 54.05 metres.

c The army wants to build another fitness station exactly the same but only has a 41 metre steel cylinder it can use for construction.
Determine how many steel cylinder steps can be made.

$a = 0.7$

$T_{23} = 0.7 + 22d$

$22d = 4.0 - 0.7$

$d = 0.15$

$\therefore T_{13} = 0.7 + (13-1)0.15 = 2.5$ metres

$S_{23} = \left(\frac{n}{2}\right)(a + a_n)$

$= \left(\frac{23}{2}\right)(0.7 + 4.0)$

= 54.05 metres ... as required

$S_n = \left(\frac{n}{2}\right)(2a + (n-1)d)$

$41 = \left(\frac{n}{2}\right)[2(0.7) + (n-1)0.15]$

$82 = 1.4n + 0.15n^2 - 0.15n$

$0 = 0.15n^2 + 1.25n - 82$

$n = \frac{-1.25 \pm \sqrt{1.25^2 + 4(0.15)(82)}}{2(0.15)}$

$= 19.58\ldots (n > 0)$

$\therefore$ 19 cylinders can be made.

$a = 300$

$\therefore$ Distance between checkpoint 2 and 3

$= a_2 - a_1 = \big((a + 2d) - a\big)$

= 350 m

24 2018 HSC Advanced Band 5

An artist posted a song online. Each day there were $2^n + n$ downloads, where n is the number of days after the song was posted.

a Find the number of downloads on each of the first 3 days after the song was posted.

b What is the total number of times the song was downloaded in the first 20 days after it was posted?

Day 1: $2^1 + 1 = 3$

Day 2: $2^2 + 2 = 6$

Day 3: $2^3 + 3 = 11$

Total downloads (20 days)

$= 2^1 + 1 + 2^2 + 2 + \ldots + 2^{20} + 20$

$= 2^1 + 2^2 + \ldots + 2^{20} + 1 + 2 + \ldots + 20$

$(GP, a = 2, r = 2) \quad (AP, a = 1, d = 1)$

$= \frac{2(2^{20} - 1)}{2 - 1} + \left(\frac{20}{2}\right)(1 + 20)$

$= 2\,097\,150 + 210$

$= 2\,097\,360$

25 HSC Sample Question Advanced Band 5

A toy train track consists of a number of pieces of track which join together.

The shortest piece of the track is 15 centimetres long and each piece of track after the shortest is 2 centimetres longer than the previous piece.

The total length of the complete track is 7.35 metres.

Find the length of the longest piece of track.

Sequence is 15, 17, 19,...

AP where $a = 15$, and $d = 2$

$S_n = \frac{n}{2}[2a + (n-1)d]$

$\therefore 735 = \frac{n}{2}[2(15) + (n-1)2]$

$= \frac{n}{2}[30 + 2n - 2]$

$= n^2 + 14n$

$0 = n^2 + 14n - 735$

Using the quadratic formula

$n = \frac{-14 \pm \sqrt{14^2 - 4(1)(-735)}}{2(1)}$

$= 21 \quad (n > 0)$

$\therefore$ Longest piece of track = $a_{21} = 15 + (21-1)2$

$= 55$ cm

26 2013 HSC Advanced Band 5

Kim and Alex start jobs at the beginning of the same year. Kim's annual salary in the first year is $30,000 and increases by 5% at the beginning of each subsequent year. Alex's annual salary in the first year is $33,000, and increases by $1,500 at the beginning of each subsequent year.

a Show that in the 10th year, Kim's annual salary is higher than Alex's annual salary.

b In the first 10 years how much, in total, does Kim earn?

c Every year, Alex saves one third of her annual salary.
How many years does it take her to save $87,500?

Let K_n = Kim's salary in Year n

K_1 = a = 30 000 $r = 1.05$

$K_{10} = ar^9 = 30\,000(1.05)^9 = \$46\,539.85$

Let A_n = Alex's salary in Year n

$A_1 = a$ = 33 000, $d = 1500$

$A_{10} = a + 9d = 33\,000 + 1500(9) = \$46\,500$

$K_{10} > A_{10}$ as required

In the first 10 years, Kim earns

$$S_{10} = \frac{a(r^n - 1)}{r - 1}$$

$$= 30\,000\left(\frac{1.05^{10} - 1}{1.05 - 1}\right)$$

$$= \$377\,336.78$$

Alex's savings: 11 000 + 11 500 + 12 000

Find n such that $S_n = 87\,500$, where $a = 11000, d = 500$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$87\,500 = \frac{n}{2}[22\,000 + (n-1)500]$$

$$87\,500 = \frac{n}{2}[21\,500 + 500n]$$

$$250n^2 + 10\,750n - 87\,500 = 0$$

$$n^2 + 43n - 350 = 0$$

$$(n - 7)(n + 50) = 0$$

$$\therefore n = 7, n > 0$$

27 HSC Sample Question Advanced Band 5

There are 10 checkpoints in a 4500 metre orienteering course.
Checkpoint 1 is the start and checkpoint 10 is the finish.

The distance between successive checkpoints increases by 50 metres as each checkpoint is passed.

Calculate the distance, in metres, between checkpoint 2 and checkpoint 3.

9 intervals exist between the 10 checkpoints; note that checkpoint 1 is $n = 0$ and checkpoint 2 is $n = 1$, and so on.

Where $d = 50 \quad S_9 = 4500$

$$S_n = \frac{n}{2}[2a + (n-1)d]$$

$$4500 = \frac{9}{2}[2a + (9-1)(50)]$$

$$4500 = \frac{9}{2}[2a + 400]$$

$$9a = 2700$$

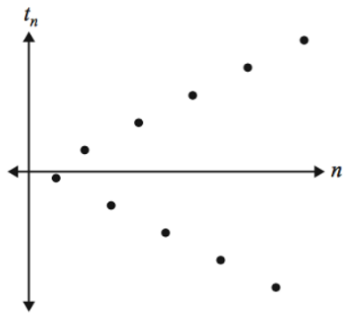
28 HSC Sample Question Advanced Band 5

The first term of a geometric sequence is a, where $a < 0$.

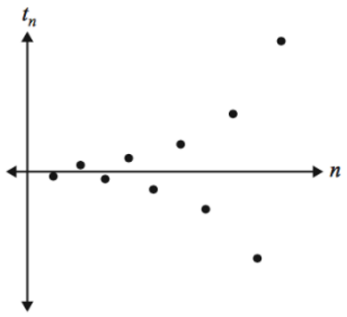
The common ratio of this sequence, r, is such that $r < -1$.

Which one of the following graphs best shows the first 10 terms of this sequence?

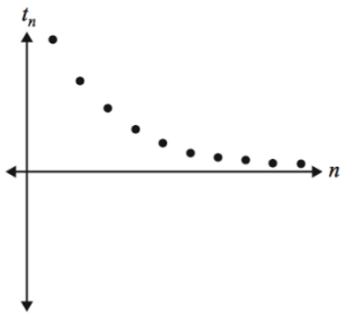
A.



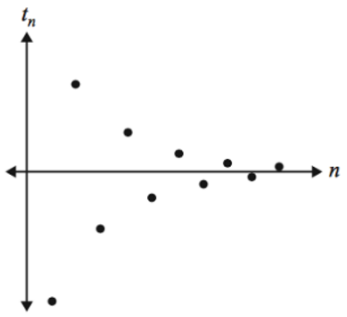
B.



C.



D.



a < 0. Cannot be C.

r < −1. Successive terms should change sign and increase exponentially with no limiting sum

Cannot be A nor D.

B is the only correct option.

29 2008 HSC Advanced Band 5

Consider the geometric series

$5 + 10x + 20x^2 + 40x^3 + \ldots$

a For what values of x does this series have a limiting sum?

$r = 2x$

$\therefore |2x| < 1$

If $2x > 0$	If $2x < 0$
$2x < 1$	$-(2x) < 1$
$x < \frac{1}{2}$	$2x > -1$
	$x > -\frac{1}{2}$

$\therefore$ Limiting sum when $-\frac{1}{2} < x < \frac{1}{2}$

b The limiting sum of this series is 100. Find the value of x.

$S_\infty = \frac{a}{1-r} = 100$

$\frac{5}{1-2x} = 100$

$100(1-2x) = 5$

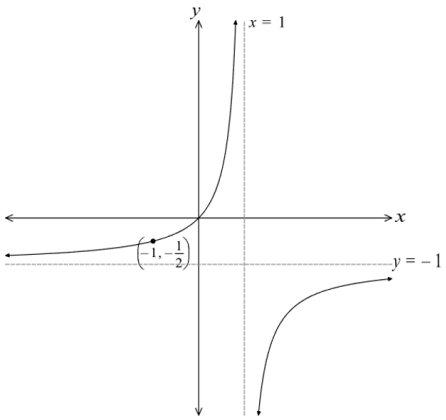
$200x = 95$

$\therefore x = \frac{95}{200} = \frac{19}{40}$

30 2024 HSC Advanced Band 5

Suppose the geometric series $x + x^2 + x^3 + \ldots$ has a limiting sum.

By considering the graph $y = -1 - \frac{1}{x-1}$, or otherwise, find the range of possible values of S_∞.



$a = x, r = x$

Limiting sum exists when $|x| < 1$

$S = \frac{a}{1-r} = \frac{x}{1-x}$

$= \frac{-x}{x-1}$

$= \frac{-x + (x-1) - (x-1)}{x-1}$

$= \frac{-x+x-1}{x-1} - \frac{x-1}{x-1}$

$= \frac{-1}{x-1} - 1$, which is the graph given.

Consider the graph, $y = S_\infty$. The domain is $|x| < 1$

$\therefore$ Domain is $-1 < x < 1$

$\therefore$ Range is $S_\infty > -\frac{1}{2}$

31 2004 HSC Advanced Band 5

Consider the geometric series $1 - \tan^2\theta + \tan^4\theta - \ldots$

a When the limiting sum exists, find its value in simplest form.

b For what values of θ in the interval $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ does the limiting sum of the series exist?

$a = 1, r = -tan^2\theta$

$\therefore S_\infty = \frac{1}{1-(-\tan^2\theta)}$

$= \frac{1}{1+\tan^2\theta}$

$= \frac{1}{\sec^2\theta}$

$= \cos^2\theta$

Find θ such that $|r| < 1$

$|-\tan^2\theta| < 1$

$\tan^2\theta < 1$

$-1 < \tan\theta < 1$

$\therefore -\frac{\pi}{4} < \theta < \frac{\pi}{4}$

32 2017 HSC Advanced Band 5

A geometric series has first term a and limiting sum 2.

Find all possible values for a.

$S_\infty = \frac{a}{1-r}$

If $S_\infty = 2$:

$2 = \frac{a}{1-r}$

$2(1-r) = a$

$1 - r = \frac{a}{2}$

$r = 1 - \frac{a}{2}$

Since $|r| < 1$,

$\left|1 - \frac{a}{2}\right| < 1$

$1 - \frac{a}{2} < 1$ or $-\left(1 - \frac{a}{2}\right) < 1$

$\frac{a}{2} > 0$ $\frac{a}{2} < 2$

$a > 0$ $a < 4$

$\therefore 0 < a < 4$

33 2011 HSC Advanced Band 6

a Rationalise the denominator in the expression $\frac{1}{\sqrt{n}+\sqrt{n+1}}$ where n is an integer and $n \geq 1$.

$$\frac{1}{\sqrt{n}+\sqrt{n+1}} \times \frac{\sqrt{n}-\sqrt{n+1}}{\sqrt{n}-\sqrt{n+1}}$$
$$= \frac{\sqrt{n}-\sqrt{n+1}}{(\sqrt{n})^2-(\sqrt{n+1})^2}$$
$$= \frac{\sqrt{n}-\sqrt{n+1}}{n-(n+1)}$$
$$= \frac{\sqrt{n}-\sqrt{n+1}}{-1}$$
$$= \sqrt{n+1}-\sqrt{n}$$

b Using your result from part (a), or otherwise, find the value of the sum

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}$$

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots+\frac{1}{\sqrt{99}+\sqrt{100}}$$
$$= (\sqrt{2}-\sqrt{1})+(\sqrt{3}-\sqrt{2})+(\sqrt{4}-\sqrt{3})+\ldots+(\sqrt{100}-\sqrt{99})$$
$$= -\sqrt{1}+\sqrt{100}$$
$$= -1+10$$
$$= 9$$

34 2013 HSC Advanced Band 6

Pat and Chandra are playing a game. They take turns throwing two dice. The game is won by the first player to throw a double six. Pat starts the game.

a Find the probability that Pat wins the game on the first throw.

$$\text{P(Pat wins on 1st throw)} = \text{P(W)}$$

$$\text{P(W)} = \text{P(Pat throws 2 sixes)}$$

$$= \left(\frac{1}{6}\right) \times \left(\frac{1}{6}\right)$$

$$= \frac{1}{36}$$

b What is the probability that Pat wins the game on the first or on the second throw?

$$\text{Let P(L)} = \text{P(loss for either player on a throw)} = \frac{35}{36}$$

$$\text{P(Pat wins on 1st or 2nd throw)}$$

$$= \text{P(W)} + \text{P(LLW)}$$

$$= \frac{1}{36} + \left(\frac{35}{36}\right) \times \left(\frac{35}{36}\right) \times \left(\frac{1}{36}\right)$$

$$= \frac{2521}{46\,656}$$

$$= 0.054 \text{ (to 3 d.p.)}$$

c Find the probability that Pat eventually wins the game.

$$\text{P(Pat wins eventually)}$$

$$= \text{P(W)} + \text{P(LL W)} + \text{P(LL LL W)} + \ldots$$

$$= \frac{1}{36} + \left(\frac{35}{36}\right)^2 \left(\frac{1}{36}\right) + \left(\frac{35}{36}\right)^2 \left(\frac{35}{36}\right)^2 \left(\frac{1}{36}\right) +$$

$$\text{Geometric series where a} = \frac{1}{36}, \text{r} = \left(\frac{35}{36}\right)^2 = \frac{1225}{1296}$$

$$\text{Since } |\text{r}| < 1\text{:}$$

$$S_\infty = \frac{a}{1 - r}$$

$$= \frac{\frac{1}{36}}{1 - \left(\frac{1225}{1296}\right)}$$

$$= \frac{36}{71}$$